

The replication principle revisited: a shared functional organization between pulvinar-cortical and cortico-cortical connectivity and its structural and molecular imaging correlates

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eLife Assessment

This study presents a **useful** characterisation of the topographical organisation of the human pulvinar, an associative thalamic subregion crucial for visual perception and attention. The evidence supporting the conclusions is **solid** given the multimodal validation and replication across datasets, although even higher-resolution imaging data would have strengthened the study. In their revised manuscript, the authors elaborated further on the motivation for their study and conducted several robustness checks. Nevertheless, there remains an opportunity for a more fully integrated interpretation of the findings. The work would be of interest to neuroscientists, neurologists, and neuropsychiatrists working on pulvinar functioning in health and disease.

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Abstract

The pulvinar, the largest nucleus in the human thalamus, is a complex, highly interconnected structure. Through a dense, organized network of cortical and subcortical areas, it provides

adequate cooperation between neural systems, which is crucial for multiple high-order functions such as perception, visuospatial attention, and emotional processing. Such a central role is made possible by a precise internal topographical organization, which is mirrored by anatomical connections as well as by the expression of neurochemical markers. While being traditionally subdivided into sub-nuclei, each characterized by distinct connectional and morphological features, recent studies in both primate and human brains have highlighted that this topographical organization only marginally aligns with the conventional histological subdivision. Instead, it has been delineated in the context of continuous gradients of cortical connections along the dorsoventral and mediolateral axes. While this multi-gradient organization has been extensively documented in primate models, it remains relatively underexplored in the human brain. The present work combines high-quality, multi-modal structural and functional imaging data with a recently published whole-brain, large-scale, positron emission tomography (PET) atlas detailing 19 neurotransmitters and receptors distributed across the human brain. By applying diffusion embedding analysis to tractography, functional connectivity, and receptor coexpression data, we identify and characterize multiple topographically organized gradients of structural connections, functional coactivation, and molecular binding patterns. We demonstrate that such gradients converge on a shared representation along the dorsoventral and mediolateral axes of the human pulvinar. This representation aligns with transitions in both structural and functional connectivity, spanning from lower-level to higher-order cortical regions. Moreover, it is paralleled by gradual changes in the expression of molecular markers associated with key neuromodulator systems, including serotonergic, noradrenergic, dopaminergic, and opioid systems. Our findings represent a significant step forward in understanding pulvinar anatomy and function, offering an exploratory framework to investigate the role of this structure in both health and disease.

Introduction

The pulvinar complex stands out as the largest nucleus within the human thalamus, serving as a pivotal hub in a myriad of cortico-subcortical networks that intricately interconnect various cortical regions of the brain (Benarroch 2015 [↗](#)). Anatomical studies in primate brains have highlighted its primary sources of afferent and efferent connections, primarily stemming from the primary and secondary visual areas. However, its connectivity also extends to other crucial regions such as the temporal lobe, primary sensory, prefrontal, and cingulate cortices (Pons and Kaas 1985 [↗](#); Romanski et al. 1997b [↗](#); Lyon and Kaas 2002 [↗](#); Fang et al. 2006 [↗](#)). Through its extensive interplay with cerebral cortical areas, the pulvinar is believed to play a pivotal role in orchestrating integrative processes crucial for context-dependent modulation of visuospatial attention (Jaramillo et al. 2019 [↗](#); Fiebelkorn and Kastner 2020 [↗](#)). It has been postulated that the pulvinar mediates the selection of relevant information from the environment by generating alpha oscillations thereby potentially modulating neuronal gain in thalamocortical circuits (Bourgeois et al. 2020 [↗](#)). Nonetheless, despite the acknowledged significance of cortico-pulvinar connectivity in these theoretical frameworks, the functional and anatomical organization of this network in the human brain remains relatively underexplored.

Traditionally, the primate pulvinar complex has been anatomically divided into four sub-nuclei: the anterior (oral), the inferior, the lateral, and the medial pulvinar nuclei (Olszewski and Baxter 1982 [↗](#)). While this subdivision has long served as a foundational framework for understanding pulvinar organization, anatomical investigations in non-human primates have revealed that connectivity patterns extend beyond discrete cytoarchitectonic subdivisions (Lysakowski et al. 1988 [↗](#); Webster et al. 1993 [↗](#); Adams et al. 2000 [↗](#)). Instead, accumulating evidence suggests a spatially continuous organization in space, wherein multiple representations of the cortical sheet and visual fields coexist (Shipp 2001 [↗](#)). These representations, often termed “maps”, adhere to the “replication principle,” wherein densely connected cortical regions exhibit overlapping

representations within the pulvinar (Shipp 2003 [↗](#)). Moreover, these representations are loosely associated with chemoarchitectural domains (Gutierrez and Cusick 1997 [↗](#); Stepniewska and Kaas 1997 [↗](#); Gutierrez et al. 2000 [↗](#)).

Translating our understanding of the functional and anatomical organization of the pulvinar complex to the human brain has proven challenging due to the small size of the underlying structures and the inherent limitations of imaging techniques. Consequently, there has been a relative scarcity of experimental works investigating pulvinar structure and function in vivo in the human brain. While functional and structural neuroimaging techniques have been successfully employed to explore the anatomy and connectivity of the human pulvinar (Leh et al. 2008 [↗](#); Tamietto et al. 2012 [↗](#); Arcaro et al. 2015 [↗](#), 2018 [↗](#); Mai and Majtanik 2019 [↗](#); Basile et al. 2021 [↗](#)), only a few studies have delved into the connectional topography of this structure. Functional MRI-based parcellation studies, employing task-based activation or resting-state connectivity fingerprinting, have provided insights into functional specialization within sub-regions of the human pulvinar (Barron et al. 2015 [↗](#); Arcaro et al. 2018 [↗](#); Guedj and Vuilleumier 2020 [↗](#)). These investigations confirm that, akin to findings in non-human primates, functional regions within the human pulvinar are loosely associated with the anatomical subdivision into nuclei. However, relying on voxel-wise clustering methods, these studies share the common assumption of discrete connectivity units and therefore may not fully capture the spatially continuous and transient nature of pulvinar connectivity patterns. Recently, gradient mapping techniques have emerged as a valuable tool for investigating the structure-function relationship in the human brain. These techniques utilize dimensional decomposition algorithms, such as diffusion embedding (Coifman and Lafon 2006 [↗](#)), to map high-dimensional to manifold low-dimensional brain features, known as “gradients”. These gradients are commonly interpreted as representing spatial patterns of smooth transitions in biological features of interest, either within or across brain structures (Haak et al. 2018 [↗](#); Hong et al. 2020 [↗](#)). Over the past decade, gradient mapping methods have been extensively applied to various biological features, including structural and functional connectivity, structural covariance, MRI-based microstructural measures, receptor, and gene expression (Bajada et al. 2017 [↗](#); Paquola et al. 2019b [↗](#), a [↗](#), 2020 [↗](#); Valk et al. 2020 [↗](#); Vos De Wael et al. 2021 [↗](#); Hansen et al. 2022b [↗](#)). These methods have been employed to explore the functional anatomy of the cerebral cortex (Margulies et al. 2016 [↗](#)), as well as specific cortical or subcortical areas, such as the striatum, thalamus, hippocampus, and cerebellum (Guell et al. 2018 [↗](#); Przeździk et al. 2019 [↗](#); Tian et al. 2020 [↗](#); Yang et al. 2020 [↗](#); Müller et al. 2020 [↗](#); Katsumi et al. 2023 [↗](#)). In particular, the existing investigations of thalamic connectivity within the gradient framework have revealed general organizational principles within the thalamus, that are partially reflected in thalamic cytoarchitecture subdivision and have been related to core and matrix thalamic neuronal subpopulation, and to their differential contribution to large-scale connectivity networks (Yang et al. 2020 [↗](#); Müller et al. 2020 [↗](#)). However, given the remarkable functional multiplicity of the pulvinar complex, it is possible that these global spatial organization patterns only partly account for the full complexity and richness of pulvinar functional topography. With this in mind, isolating pulvinar connectivity from the remaining thalamocortical connectome would ensure that local organizational principles are not obscured by the global connectotopy of the entire thalamus.

The present study aims to investigate the gradient organization of the human pulvinar using a multimodal approach. By leveraging high-quality structural, diffusion-weighted, and resting state functional MRI from two independent datasets (Human Connectome Project; Leipzig Mind-Brain-Body dataset) (Van Essen et al. 2013 [↗](#); Babayan et al. 2019 [↗](#)), along with a recently published multi-tracer receptor expression atlas derived from ~1000 positron emission tomography (PET) scans (Hansen et al. 2022b [↗](#)), we aim to provide insights on the spatial organization of connectivity within the human pulvinar and its relation to molecular expression.

Results

1. The multiscale gradient organization of the human pulvinar goes beyond discrete anatomical subdivisions

We collected functional and diffusion-weighted imaging data from two high-quality, independent datasets of healthy subjects: a primary dataset from the Human Connectome Project (HCP) (Van Essen et al. 2012 [↗](#)), including 210 healthy subjects (males=92, females=118, age range 22-36 years); and a validation dataset from the Leipzig Mind-Brain-Body dataset (LEMON) (Babayan et al. 2019 [↗](#)), consisting of 213 healthy subjects (males=138, females=75, age range 20-70 years). The pulvinar complex, along with its constituent subnuclei, was delineated bilaterally in the left and right hemispheres according to a whole brain labeling atlas (Rolls et al. 2020 [↗](#)). For each participant, we generated voxel-wise estimates of pulvinar functional and structural connectivity to a 400-label parcellation of the cerebral cortex (Schaefer et al. 2018 [↗](#)), respectively from preprocessed resting-state functional MRI and from whole brain probabilistic tractograms derived from diffusion data. Individual connectivity maps were then normalized and aggregated to group-level, dense functional and structural connectomes.

From the normative neurotransmitter atlas (Hansen et al. 2022b [↗](#)), we extracted the voxel-wise density profiles of 19 receptors and transporters across nine different neurotransmitter systems (serotonin, noradrenaline, dopamine, glutamate, gamma-aminobutyric acid, acetylcholine, histamine, opiates, endocannabinoids), and we calculated the pairwise correlation for each voxel (coexpression). The functional and structural dense connectomes, as well as the coexpression matrix, were converted to modality-specific affinity matrices quantifying the similarity of each voxel profile to each other.

By applying diffusion embedding (Coifman and Lafon 2006 [↗](#)) on modality-specific affinity matrices sampled from left and right pulvinar, we derived distinct functional connectivity, structural connectivity, and receptor expression embeddings (**Figure 1** [↗](#)).

These embeddings, which we henceforth refer to as “gradients”, represent unitless entities in which each value identifies the position of pulvinar voxels along the respective embedding axis that encodes dominant differences in voxel connectivity or coexpression patterns (Margulies et al. 2016 [↗](#); Haak et al. 2018 [↗](#)).

The first three functional connectivity gradients (G_{FC1} - G_{FC3}) collectively explained ~80% of the total variance in functional connectivity for both the left and right pulvinar, as evidenced by the elbow observed in the scree plot (**Figure 2A** [↗](#)). The principal gradient (G_{FC1} , ~50% of total variance explained for both left and right pulvinar) reflected a dorsomedial-to-ventrolateral topographical organization. Of the two secondary gradients, explaining each approximately ~10/15% of the total variance for both left and the right pulvinar, one (G_{FC2}) ran to the posteromedial part of the pulvinar to the most anterior dorsal region, the other (G_{FC3}) was aligned to the posterolateral to the anteromedial axis (**Figure 2A** [↗](#)).

Nearly 90% of the global variance in structural connectivity values was explained by the first two connectivity gradients (G_{SC1} - G_{SC2}). The first structural gradient (G_{SC1} , ~70% of total variance explained, bilaterally) was arranged on a medio-lateral axis, while the second (G_{SC2} , explaining approximately 20% of global explained variance, delineated a dorsal-to-ventral topography (**Figure 2B** [↗](#)).

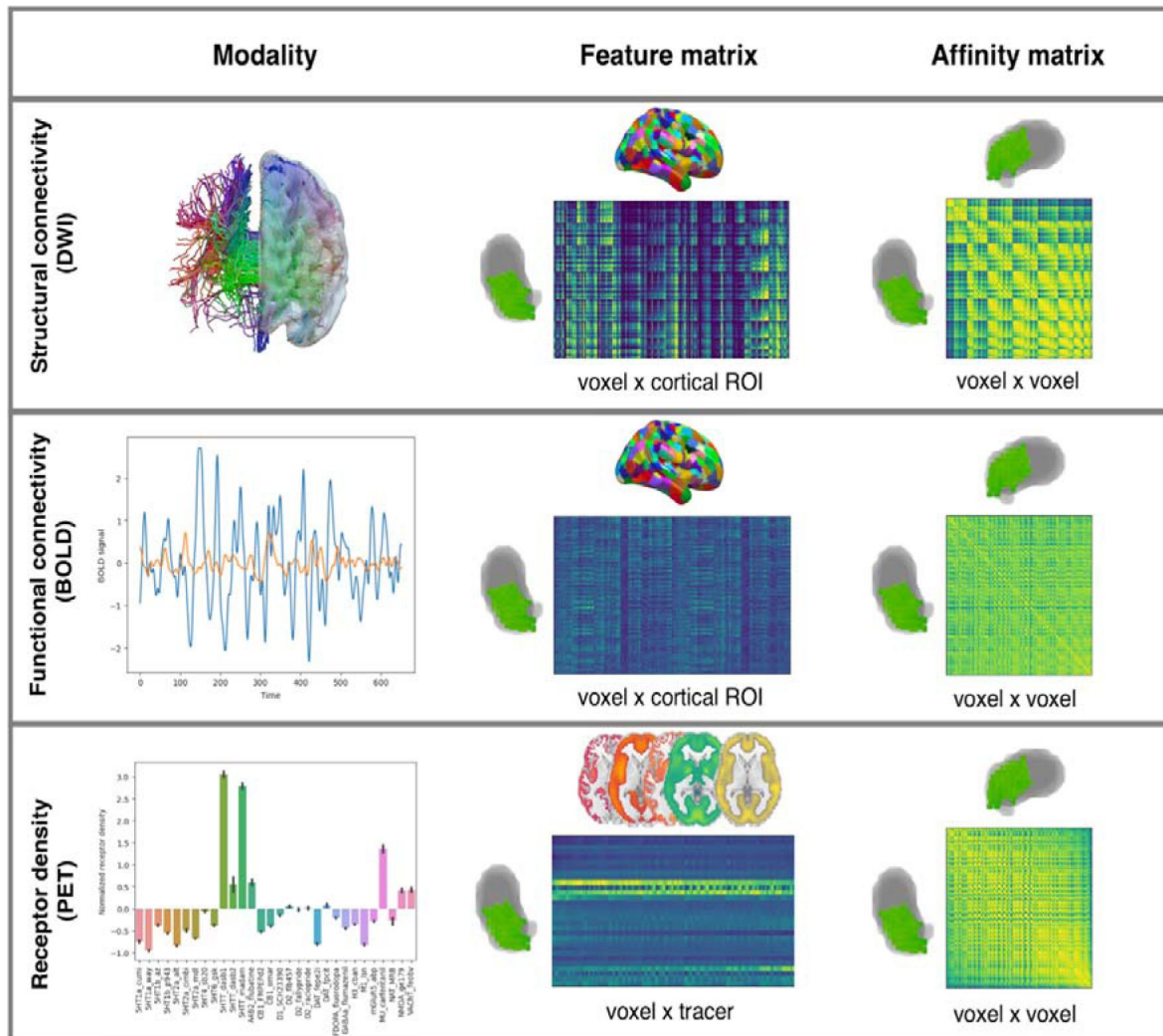


Figure 1.

Schematic overview of the gradient mapping protocol.

For each different imaging modality (DWI; BOLD; PET) a feature matrix was built by extracting voxel-wise features in the left and right pulvinar separately. Structural and functional connectivity to 400 cortical areas, as well as the expression profiles of 19 receptors and transporters, were extracted for each pulvinar voxel. Symmetric, affinity matrices comparing the feature profile similarity of each voxel against each other were estimated using cosine similarity and fed into the diffusion embedding algorithm. Additional details can be found in the main text.

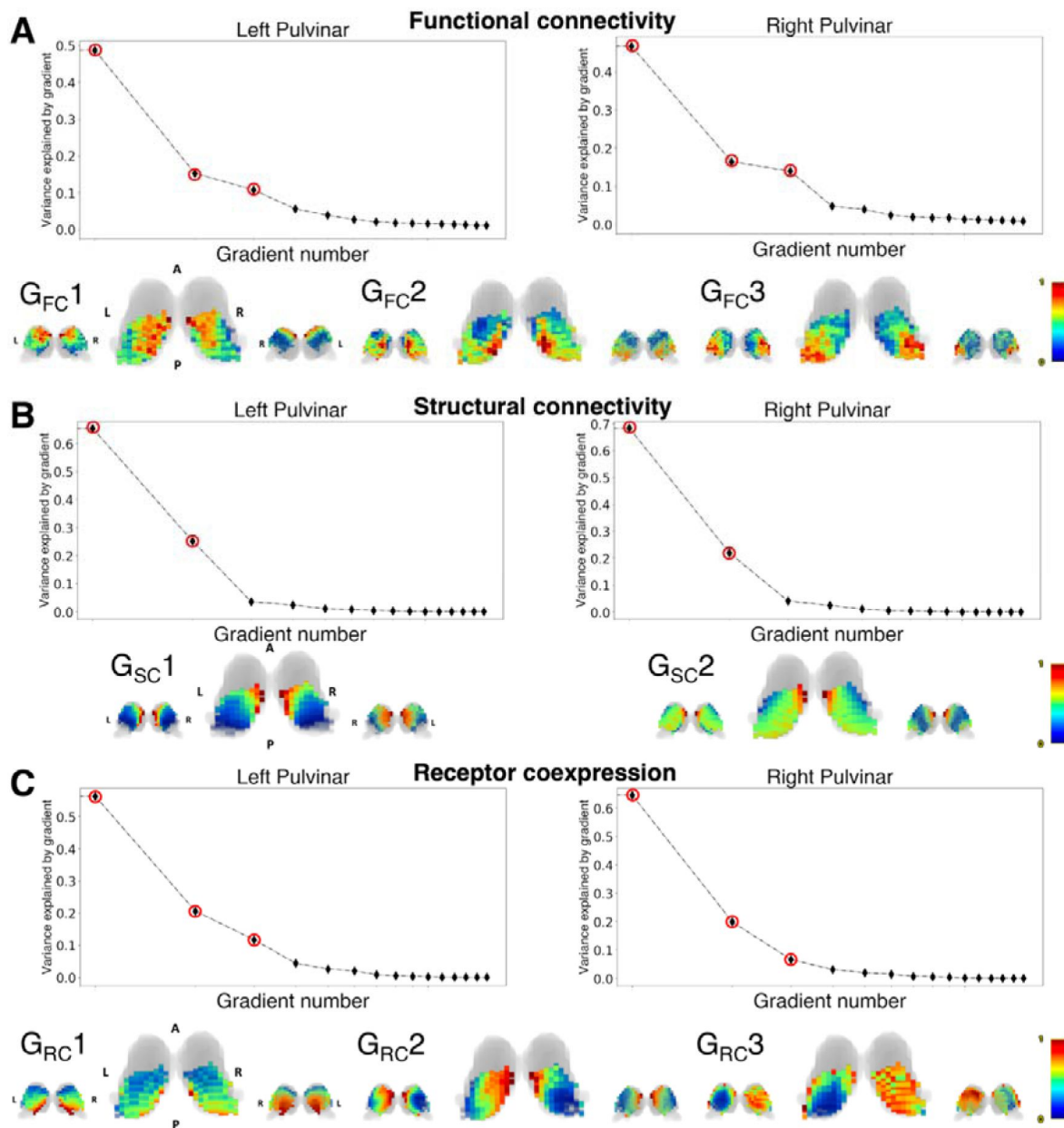


Figure 2.

Functional connectivity, structural connectivity, and receptor coexpression pulvinar gradients.

For each panel, the top row shows the scree plots of explained variance for each gradient. The gradients that have been considered for subsequent analyses are marked with red circles. The bottom row shows 3D reconstruction of the normalized gradient images overlaid on the thalamic outline (gray-shaded area). The thalamic volume has been obtained from the AAL3 atlas. Gradients are shown in posterior (left), superior (center), and anterior (right) views. A: anterior; P: posterior; L: left; R: right.

Finally, the first three gradient embeddings (G_{RC1} - G_{RC3}) explained globally ~80% of the variance in receptor coexpression data both for the left and right pulvinar, although with minor differences between the left and right hemispheres for individual gradients. The main gradient (G_{RC1}), explaining 60% of the total variance in the left hemisphere and 70% in the right, depicted a dorso-medial to ventro-lateral axis organization. The second gradient (G_{RC2}), accounting for ~20% of the explained variance on the left hemisphere and ~15% on the right hemisphere, displayed a medio-lateral topography. Finally, the third gradient (G_{RC3}), capturing approximately 10% of the total variance on the left and less than 10% on the right, ran along the anterior-posterior axis (**Figure 2C**).

Due to the small volume of the pulvinar complex, whose boundaries can vary significantly across different thalamic segmentation, we validated our results on an additional, thalamo-specific atlas (Su et al. 2019). We found that the results of the diffusion embedding are highly consistent, regardless of the imaging modality of choice, both in terms of variance explained and spatial topography (**Supplementary Figure 1**).

To characterize the relationship between these pulvinar gradients and the conventional anatomical subdivision of the pulvinar complex into discrete nuclei, we analyzed the distribution of gradient values across the four traditional anatomical nuclei (medial, lateral, anterior, inferior) as defined by an atlas-based parcellation based on post-mortem histology (Iglesias et al. 2018). First, by plotting gradient values grouped by each pulvinar nucleus, we observed limited correspondence to discrete pulvinar nuclei, regardless of the imaging modality. Although some gradients showed a progressing trend along different pulvinar nuclei, most gradient values were evenly distributed across nuclei. To assess if discrete pulvinar nuclear structure could be inferred from the continuous gradient values derived from diffusion embedding, k-means data-driven clustering in gradient space was applied, either by considering the most relevant gradients for each single imaging modality (functional, structural, or receptor coexpression) or by concatenating gradient values from all the modalities. The appropriate number of clusters (4 for functional gradients, 3 for structural gradients, 4 for coexpression gradients, and 8 for combined clustering) was selected based on silhouette score plots for each hemisphere. Clustering of gradient values resulted in generally poor overlap to anatomical nuclei either for single modality (maximum Dice coefficient left/right pulvinar: 0.52/0.49 to medial pulvinar for functional gradients, 0.72/0.66 to medial pulvinar for structural gradients; 0.47/0.46 to medial pulvinar for coexpression gradients) or concatenated modalities (maximum Dice coefficient left/right pulvinar: 0.48/0.46 to anterior pulvinar) (**Supplementary Figure 2**).

2. Pulvinar-cortical functional connectivity replicates patterns of cortico-cortical functional connectivity

Connectivity gradients serve as low-dimensional representations of the topographical organization patterns of connectivity, whether functional or structural connectivity. With this concept in mind, we sought to investigate the extent to which pulvinar-cortical connectivity reflects cortico-cortical connectivity. To achieve this, we generated pulvinar-cortical gradient-weighted connectivity maps for the most relevant structural and functional gradients. This involved calculating the average of the dot products of gradient values in each voxel and their connectivity to each cortical ROI. Subsequently, we correlated these maps with gradients derived from the embedding of cortico-cortical connectivity. To address the issue of spatial autocorrelation (SA), which commonly affects brain maps, we computed SA-corrected permutational p-values using a method based on SA-preserving surrogates (Burt et al. 2020). Furthermore, these SA-corrected p-values were adjusted for false discovery rate (FDR) using the Benjamini-Hochberg correction method.

Diffusion embedding of cortico-cortical functional connectivity gradients revealed an elbow in the explained variance graph at the fifth gradient, collectively explaining ~55% of the total variance. Similarly, the first four gradients of cortico-cortical structural connectivity accounted for ~60% of

the total variance (**Supplementary Figure 2** [↗](#)–[3](#) [↗](#)).

We found that pulvinar-cortical connectivity patterns associated with pulvinar gradients exhibited high correlations with the first three cortico-cortical functional connectivity gradients. Specifically, G_{FC1} -weighted connectivity maps demonstrated the highest correlation with the first cortico-cortical functional gradient (left pulvinar: $r=0.89$, $p < 0.01$; right pulvinar: $r=0.83$, $p < 0.01$); in line with the available evidence, this gradient delineates a unimodal-to-transmodal cortical hierarchy, spanning from low-level sensory and motor regions to higher-order associative and limbic cortices (Margulies et al. 2016 [↗](#)). G_{FC2} -weighted connectivity maps strongly correlated with the third cortico-cortical functional gradient (left pulvinar: $r=0.78$, $p < 0.01$; right pulvinar: $r=0.78$, $p < 0.01$); as in previous investigations, the cortical topography of this gradient spans across cortical regions involved in low-versus-high complexity cognitive task, thus highlighting a cortical organization depending on cognitive demand (Turnbull et al. 2020 [↗](#)). Finally, G_{FC3} -weighted connectivity was found strongly correlated with the second cortico-cortical functional gradient (left pulvinar: $r=0.71$, $p < 0.01$; right pulvinar: $r=0.67$, $p < 0.01$): in line with previous works, this cortical gradient is anchored on one end on visual processing areas, and on the opposite end on sensorimotor processing networks, reflecting a secondary cortico-cortical hierarchy that is orthogonal to the principal cortical gradient (Margulies et al. 2016 [↗](#)). Additionally, weaker yet significant correlations were found between G_{FC1} -weighted connectivity maps and the second cortico-cortical functional gradient (left pulvinar: $r=0.27$, $p=0.01$; right pulvinar: $r=0.38$, $p < 0.01$) (**Figure 3A** [↗](#)).

On the contrary, gradient-weighted structural connectivity maps did not exhibit the same correlation pattern with cortico-cortical structural gradient maps, displaying only weaker correlations to cortical gradients. In both left and right pulvinar, G_{SC2} -weighted structural connectivity maps showed weak correlations with the principal structural connectivity gradient (left pulvinar: $r=0.31$, $p=0.02$; right pulvinar: $r=0.28$, $p < 0.01$). Additionally, there was a trend towards significance observed for the correlation between left G_{SC1} -weighted connectivity and the third cortico-cortical gradient ($r=0.17$, $p=0.02$) and between right G_{SC2} -weighted connectivity and the third cortico-cortical gradient ($r=0.19$, $p=0.01$) (**Supplementary Figure 3** [↗](#)).

3. The unimodal-to-transmodal gradient (G_{FC1}) aligns with receptor expression on the dorso-ventral pulvinar axis

We discovered that the main gradient of cortico-cortical functional connectivity is mirrored on the pulvinar by G_{FC1} . The cortical connectivity of this gradient progresses from unimodal cortical areas of the sensorimotor and visual network to multi-modal, higher-order associative areas of the limbic, frontoparietal, and default mode network (**Figure 3B** [↗](#)). Similarly, in the pulvinar, this trend manifests as a progression of gradient values in a ventral-to-dorsal fashion, from the anterior and inferior to the lateral and medial pulvinar nuclei (**Figure 4B** [↗](#)).

We further delved into the relationship between multi-scale organization features of the pulvinar by correlating gradients obtained from embedding data obtained from different imaging modalities. To take into account the effects of spatial autocorrelation, we corrected the resulting p-values using a method based on SA-preserving spatial null models (Burt et al. 2020 [↗](#)). For this gradient, we found the strongest association with the main gradient of receptor expression G_{RC1} (**Figure 4A** [↗](#); left pulvinar: $r=-0.71$, $p < 0.01$; right pulvinar: $r=-0.78$, $p < 0.01$). It is worth noting that, due to the intrinsic sign indeterminacy of gradient values, absolute correlation values were considered. Similarly to G_{FC1} , G_{RC1} was also organized along the dorso-ventral axis of the pulvinar complex and displayed a similar distribution of values across pulvinar nuclei, albeit inverted with respect to G_{FC1} , with loadings progressing from lateral and medial to anterior and inferior pulvinar nuclei.

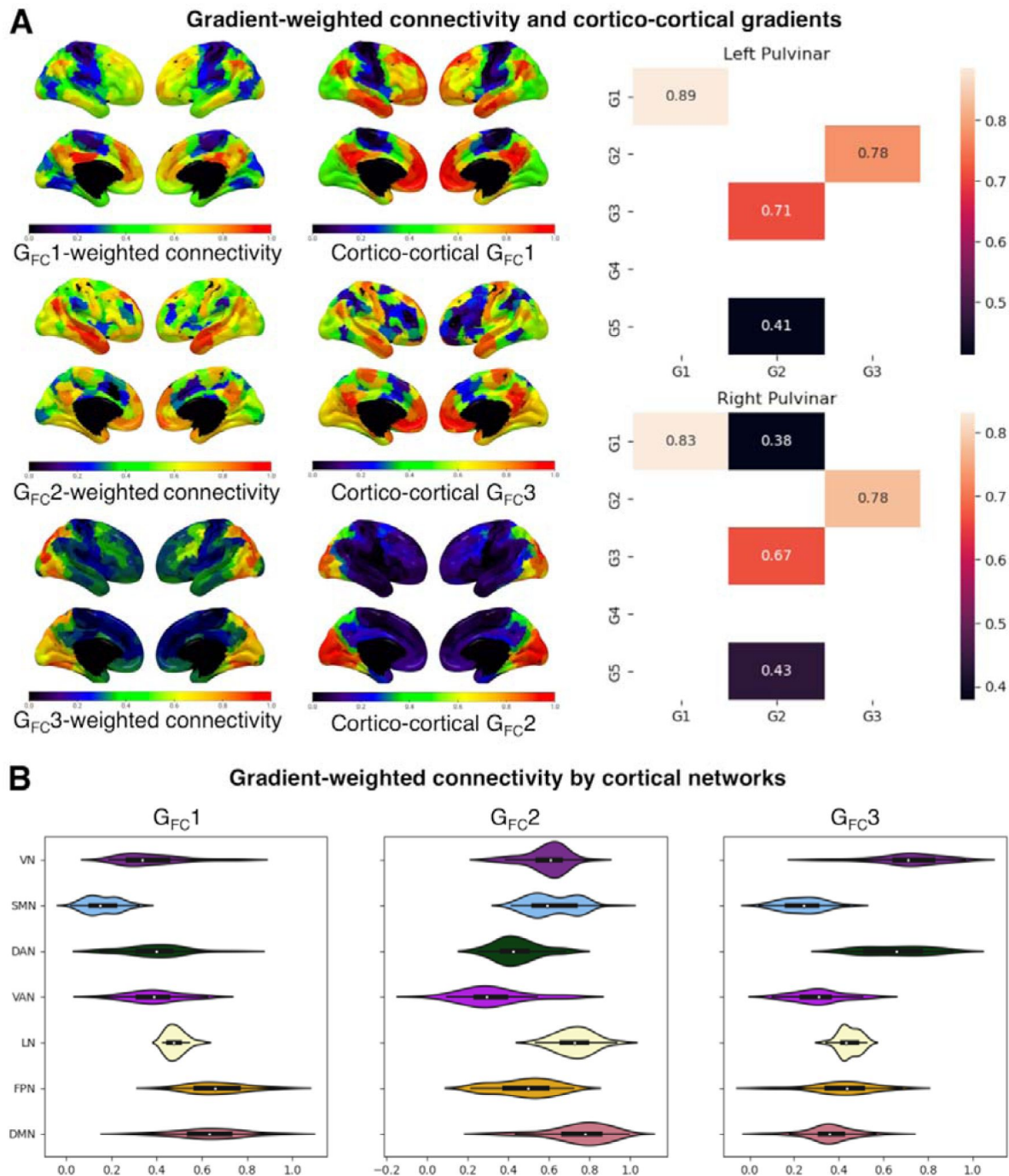


Figure 3.

A) The relationship between pulvinar-cortical and cortico-cortical functional gradients. The left panels show gradient-weighted connectivity values for the first three cortico-pulvinar functional connectivity gradients, each paired with the most correlated cortico-cortical functional connectivity gradient. Normalized values of each cortical ROI (Schaefer atlas, 400 parcels) are plotted on the cortical surface. The right panels show matrix plots of Pearson's correlation values between cortico-cortical gradients (y-axis) and pulvinar-cortical gradient-weighted connectivity maps. Only values showing statistical significance (SA-corrected, FDR-adjusted $p < 0.01$) are shown. **B) The relationship between pulvinar-cortical functional gradients and cortical connectivity networks.** Violin plots of normalized gradient-weighted connectivity values grouped by seven intrinsic connectivity networks (as in Yeo et al. 2011 [4]); VN: visual network; SMN: sensorimotor network; DAN: dorsal attention network; VAN: ventral attention network; LN: limbic network; FPN: frontoparietal network, DMN: default-mode network.

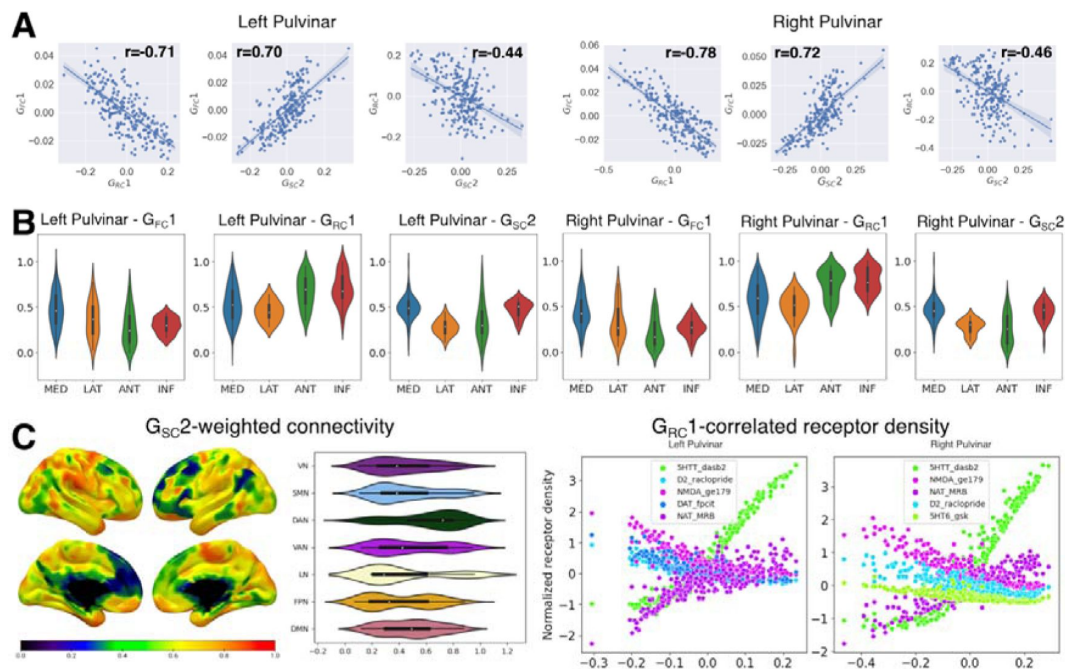


Figure 4.

A) Pulvinar correlates of the unimodal-transmodal cortical gradient. Scatter plots illustrate the relationship between the first pulvinar-cortical connectivity gradient, corresponding to the unimodal-transmodal hierarchy of cortico-cortical connectivity, and its most correlated gradient values across other modalities. Due to the intrinsic sign indeterminacy of gradient values, absolute correlation values are considered. **B) The relationship between correlated pulvinar gradients and discrete histological nuclei.** Violin plots illustrating normalized gradient values grouped by histological nuclei (AAL atlas). MED: medial pulvinar; LAT: lateral pulvinar; ANT: anterior pulvinar; INF: inferior pulvinar. **C) Structural connectivity and receptor coexpression patterns correlating with the unimodal-transmodal hierarchy.** Left panels: gradient-weighted structural connectivity. Normalized values for each cortical ROI (Schaefer atlas, 400 parcels) are plotted on the cortical surface. Violin plots show values grouped by seven intrinsic connectivity networks (as in Yeo et al. 2011). VN: visual network; SMN: sensorimotor network; DAN: dorsal attention network; VAN: ventral attention network; LN: limbic network; FPN: frontoparietal network; DMN: default-mode network. Right panels: gradient-correlated receptor density values for the top 5 most correlated receptors. Details can be found in the main text.

To identify the receptor coexpression patterns driving this specific mode of organization, we examined the values of the five tracers with the strongest correlation to gradient values. We observed that expression of the serotonin and noradrenaline reuptake transporters (5HTT and NAT) positively correlated with gradient values. Conversely, the expression of markers of dopaminergic activity, such as the D2 receptor or the reuptake transporter DAT, as well as NMDA glutamatergic receptors decreased with increasing gradient values (**Figure 4C** [↗](#)).

Furthermore, we observed a significant correlation between G_{FC1} and the secondary gradient of structural connectivity G_{SC2} (left pulvinar: $r=0.70$, $p < 0.01$; right pulvinar: $r=0.72$, $p < 0.01$), which explained $\sim 20\%$ of the variance in structural connectivity. Similar to G_{FC1} , G_{SC2} follows the dorso-ventral axis of the pulvinar complex, progressing from lower values in the anterior and lateral nuclei to higher values in the inferior and medial nuclei (**Figure 4C** [↗](#)). Consequently, this gradient also displayed a weaker negative correlation with G_{RC1} (left pulvinar: $r=-0.44$, $p < 0.01$; right pulvinar: $r=-0.46$, $p < 0.01$). However, the cortical structural connectivity pattern associated with G_{FC1} does not mirror that of its functional connectivity counterpart. Instead, gradient-weighted structural connectivity progresses from lower values in modality-independent networks (e.g., default mode, frontoparietal) to intermediate values in unimodal, low-level sensory networks (e.g., visual and sensorimotor) with maximal values in networks involved in higher-order sensory processing (e.g., limbic, ventral attention, dorsal attention).

In summary, our findings demonstrated that G_{FC1} , G_{RC1} , and G_{SC2} substantially delineate multiscale differences between the ventral and dorsal aspects of the pulvinar. Moving along the ventral-dorsal axis of the pulvinar complex, more ventral regions showed higher functional connectivity to unimodal sensory processing networks, higher levels of 5HTT and NAT expression, and preferentially higher structural connectivity to modality-independent or low-level sensory processing cortices. Conversely, dorsal regions displayed increasingly higher functional connectivity to multi-modal processing networks, higher levels of D2, NMDA receptors, and greater structural connectivity to higher-order sensory processing cortices (**Figure 4** [↗](#)).

4. The visual-to-sensorimotor gradient (G_{FC3}) aligns with structural connectivity on the medio-lateral pulvinar axis

The secondary gradient of cortico-cortical organization was found to be correlated with G_{FC3} , which explained $\sim 10\%$ of the variance in pulvinar-cortical functional connectivity. G_{FC3} exhibited a well-delineated medial-lateral and antero-posterior arrangement, corresponding to lower values in the anterior and lateral pulvinar divisions, and higher values in the inferior and medial nuclei. The cortical functional connectivity patterns associated with G_{FC3} spanned from the lower extreme of the sensorimotor network to the higher extreme of the visual network. This gradient reflected transitions from increasingly visual-related processing networks (such as dorsal attention and limbic networks) to increasingly action-related processing networks (including the default mode, frontoparietal, and ventral attention networks) (**Figure 5B** [↗](#)).

We found that G_{FC3} is the most correlated, in absolute values, to the principal gradient G_{SC1} , which explained $\sim 70\%$ of variance in pulvinar-cortical structural connectivity, showing a strong negative correlation (**Figure 5A** [↗](#); left pulvinar: $r=-0.61$, $p < 0.01$; right pulvinar: $r=-0.77$, $p < 0.01$). G_{SC1} also delineates a clear progression along the medio-lateral axis of the pulvinar, with lower values in the inferior and medial pulvinar nuclei transitioning to higher values in the lateral and anterior pulvinar nuclei (**Figure 5B** [↗](#)). On the cortical mantle, the structural gradient-weighted connectivity patterns exhibited a heterogeneous distribution of values within functional networks, rather than delineating a specific pattern of progression across functional networks (**Figure 5C** [↗](#)).

Additionally, we found a significant negative correlation to G_{RC2} , which accounts for $\sim 15\%$ of the variance in receptor expression (left pulvinar: $r=-0.47$, $p < 0.01$; right pulvinar: $r=-0.72$, $p < 0.01$). This gradient consistently showed an association in both hemispheres with the expression of mu-

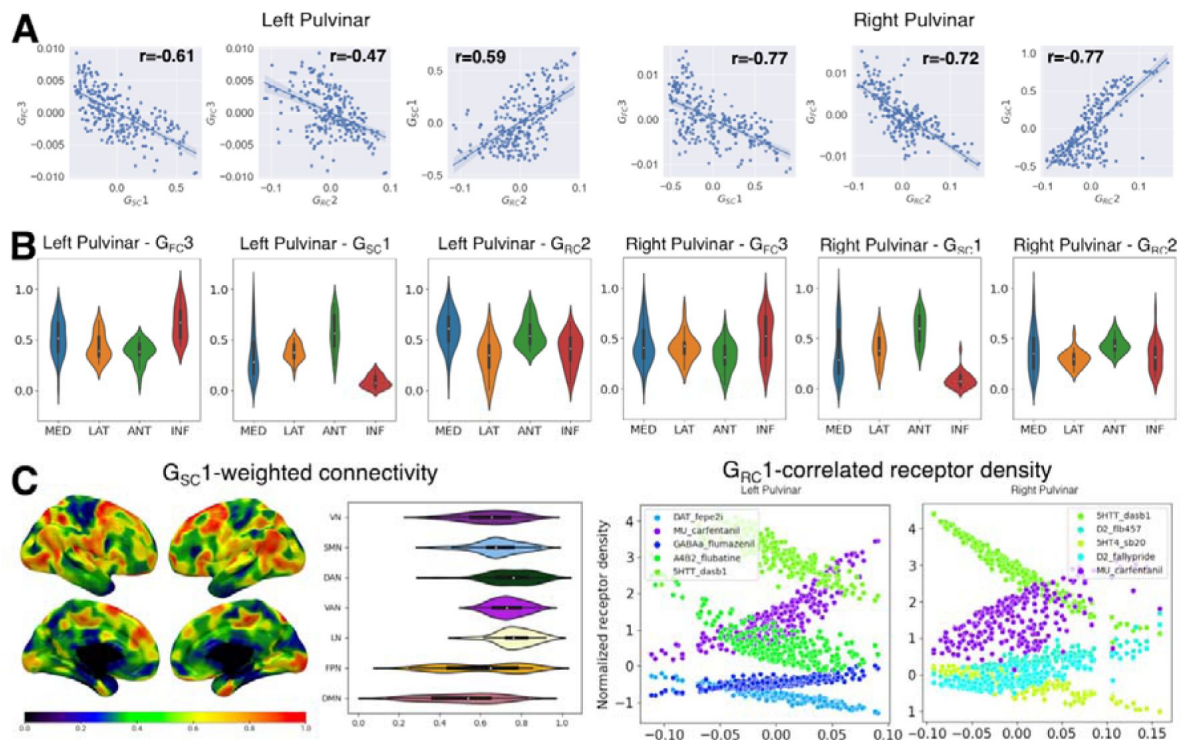


Figure 5

A) Pulvinar correlates of the visual-to-sensorimotor cortical gradient. Scatter plots illustrate the relationship between the third pulvinar-cortical connectivity gradient, corresponding to the visual-to-sensorimotor hierarchy of cortico-cortical connectivity, and its most correlated gradient values across other modalities. Due to the intrinsic sign indeterminacy of gradient values, absolute correlation values are considered. **B) The relationship between correlated pulvinar gradients and discrete histological nuclei.** Violin plots illustrating normalized gradient values grouped by histological nuclei (AAL atlas). MED: medial pulvinar; LAT: lateral pulvinar; ANT: anterior pulvinar; INF: inferior pulvinar. **C) Structural connectivity and receptor coexpression patterns correlating with the visual-to-sensorimotor hierarchy.** Left panels: gradient-weighted structural connectivity. Normalized values for each cortical ROI (Schaefer atlas, 400 parcels) are plotted on the cortical surface. Violin plots show values grouped by seven intrinsic connectivity networks (as in Yeo et al. 2011). VN: visual network; SMN: sensorimotor network; DAN: dorsal attention network; VAN: ventral attention network; LN: limbic network; FPN: frontoparietal network, DMN: default-mode network. Right panels: gradient-correlated receptor density values for the top 5 most correlated receptors. Details can be found in the main text.

opioid receptors (MOR), which was positively correlated to the gradient axis, and of the 5HTT, which was negatively correlated to the gradient axis (**Figure 5C**). Similar to G_{FC3} , G_{RC2} was also found to be correlated with the main gradient of structural connectivity G_{SC1} (**Figure 5A**; left pulvinar: $r = 0.59$, $p < 0.01$; right pulvinar: $r = 0.77$, $p < 0.01$). Like G_{SC1} , G_{RC2} exhibited a progression of values from inferior and medial pulvinar nuclei to the lateral and anterior nuclei (**Figure 5B**).

To summarize, our findings reveal a strong association between the medio-lateral axis of pulvinar organization, which reflects the progression of functional connectivity from sensorimotor to associative and visual areas, and the principal gradient of structural connectivity. The cortical connectivity profiles of this gradient delineate within-network patterns of progression, suggesting that multiple functional networks host separate representations of the pulvinar complex. Additionally, this structural and functional connectivity pattern is paralleled by a secondary gradient of receptor expression. Specifically, there is increasing expression of MOR and decreasing expression of 5HTT in the medial end of the gradient, which corresponds to regions functionally connected to sensorimotor-associative regions. Conversely, there is a decreasing expression of MOR and an increasing expression of 5HTT in the lateral end of the gradient, which corresponds to regions functionally connected to associative-visual regions (**Figure 5**).

5. Reliability and reproducibility

In general, diffusion embedding of cortico-pulvinar connectivity data demonstrated high stability and reproducibility, with structural connectivity gradients exhibiting superior performance compared to functional connectivity gradients.

The structural connectivity gradients G_{SC1} and G_{SC2} showed excellent stability across 100 resampling of random halves of the main dataset in both hemispheres (all median $r > 0.99$ for both left and right pulvinar, with $IQR < 0.01$). Additionally, group-level structural gradients demonstrated the highest repeatability when compared between test and retest samples of the main dataset (all $r > 0.99$ for both left and right pulvinar). Finally, structural gradients showed high reproducibility across both the main and validation datasets (left pulvinar, G_{SC1} : $r = 0.87$; G_{SC2} : $r = 0.88$; right pulvinar, G_{SC1} : $r = 0.87$; G_{SC2} : $r = 0.85$).

Functional connectivity gradients showed overall very high or moderate-to-high stability across split-half resamples of the main dataset in both hemispheres. However, stability decreased with decreasing variance explained (left pulvinar, G_{FC1} : median $r = 0.85$, $IQR = 0.021$; G_{FC2} : median $r = 0.70$, $IQR = 0.042$; G_{FC3} : median $r = 0.59$, $IQR = 0.061$; right pulvinar, G_{FC1} : median $r = 0.87$, $IQR = 0.023$; G_{FC2} : median $r = 0.75$, $IQR = 0.039$; G_{FC3} : median $r = 0.72$, $IQR = 0.039$).

Repeatability on test-retest samples of the main dataset also showed a decreasing trend following the decrease of explained variance, with moderate to good repeatability for G_{FC1} and G_{FC2} , and low repeatability for G_{FC3} . In contrast, reproducibility of the group-level gradients across the main and validation sample, while showing a similar trend with higher values for G_{FC1} compared to the other secondary gradients, was found high both for left (all $r > 0.75$) and right pulvinar (all $r > 0.80$).

To assess the influence of local noise on functional and structural connectivity gradients, we calculated the spatial correlation between gradient values and averaged voxel-wise estimates of signal-to-noise ratio (SNR) from functional and structural MRI data, respectively. We found that functional connectivity gradients are weakly, but significantly correlated with the SNR, with the strongest correlation observed for the third gradient (left hemisphere G_{FC1} $r = -0.30$, SA-corrected $p < 0.001$, G_{FC2} $r = 0.22$, SA-corrected $p = 0.05$, G_{FC3} $r = 0.55$, SA-corrected $p < 0.001$; right hemisphere G_{FC1} $r = -0.41$, SA-corrected $p < 0.001$, G_{FC2} $r = 0.22$, SA-corrected $p = 0.008$, G_{FC3} $r = 0.52$, SA-corrected $p = 0.017$). In contrast, structural connectivity gradients were not significantly associated

with SNR (left hemisphere G_{SC1} $r = 0.06$, SA-corrected $p = 0.82$, G_{SC2} $r = -0.33$, SA-corrected $p = 0.01$; right hemisphere G_{SC1} $r = 0.40$, SA-corrected $p = 0.28$, G_{SC2} $r = -0.19$, SA-corrected $p = 0.31$) (Supplementary Figure 5 [↗](#)).

Discussion

In the present study, we employed a data-driven dimensionality reduction analysis on a comprehensive array of high-resolution, large-scale multimodal datasets encompassing structural, functional connectivity, and receptor expression data. Our objective was to delve into the organizational principles underlying pulvinar-cortical connectivity and its relationship to cortico-cortical connectivity. Through the application of diffusion embedding, a widely utilized gradient analysis technique, we successfully collapsed the high-dimensional connectivity and coexpression data into a reduced set of low-level representations, reflecting the spatial patterns of organization of multimodal features on the pulvinar complex.

Previous reports on functional or structural connectivity of the human pulvinar complex have been predominantly either ROI-based and hypothesis-driven ([Leh et al. 2008](#) [↗](#); [Tamietto et al. 2012](#) [↗](#); [Arcaro et al. 2015](#) [↗](#); [Basile et al. 2021](#) [↗](#)) and thus unable to capture finer details regarding the topographical organization patterns of pulvinar connectivity in its entirety, or data-driven and focused on hard parcellations from data clustering algorithms ([Barron et al. 2015](#) [↗](#); [Guedj and Vuilleumier 2020](#) [↗](#)). The later methodological approach tends to force the topographical heterogeneity of pulvinar connectivity patterns into discrete, spherical units rather than providing insights into their continuous transitions in space.

Over the past five decades, the connectivity of the pulvinar complex has undergone through investigations in non-human primates, utilizing anatomical tract-tracing or electrophysiological methods ([Asanuma et al. 1985](#) [↗](#); [Baleydier and Mauguier 1985](#) [↗](#); [Pons and Kaas 1985](#) [↗](#); [Cusick and Gould 1990](#) [↗](#); [Baleydier and Morel 1992](#) [↗](#); [Romanski et al. 1997a](#) [↗](#); [Shipp 2001](#) [↗](#)). A prevailing principle suggests that connectivity to cortical regions does not adhere to clear-cut distinctions between histological subdivisions of the pulvinar complex but instead follows a topographical organization spanning across anatomical nuclei ([Shipp 2003](#) [↗](#)). Accordingly, existing connectivity-based parcellation models of the pulvinar complex have generally shown moderate-to-low correspondence between functional connectivity or coactivation clusters and histological delineations of pulvinar nuclei ([Barron et al. 2015](#) [↗](#); [Ji et al. 2016](#) [↗](#); [Guedj and Vuilleumier 2020](#) [↗](#)). Our results align with this perspective by demonstrating that while structural and functional connective topography often follows recognizable patterns of progression across anatomical nuclei, hard clustering of either connectivity or coexpression data failed to reveal the underlying boundaries of nuclear anatomy. In other words, the topographical pattern of connection and coexpression only partially reflects the anatomical boundaries outlined by the connectivity or coexpression differences between nuclei.

It is worth noting, however, that connectivity profiles of individual pulvinar nuclei can be properly inferred from gradient values and their relative gradient-weighted connectivity maps. For instance, G_{FC1} , which showed progression of gradient-weighted connectivity from low-level sensorimotor and visual processing regions to higher-level multimodal, limbic, and default-mode regions, demonstrated a corresponding progression of values from anterior (mostly sensorimotor) and inferior (low-level visual) nuclei to lateral (higher-level visual) and medial (associative and limbic) pulvinar nuclei. This observation is consistent with existing literature in non-human primates ([Benarroch 2015](#) [↗](#); [Homman-Ludiyé and Bourne 2019](#) [↗](#)). In synthesis, the notion of connectivity gradients reconciles the concept of continuous and graded cortical representations on the pulvinar complex with the hypothesis of functional and anatomical specialization of discrete pulvinar subregions.

Drawing from anatomical literature in primates, Shipp (2003) [proposed a model of pulvinar-cortical connectivity](#), positing a clear dissociation between dorsal and ventral pulvinar, with each region hosting distinct and parallel representations of cortical regions along their latero-medial axis. According to this model, connections to the dorsal pulvinar are organized along a dorsal parietal-superior temporal axis, while connections to the ventral pulvinar follow an occipital-inferior temporal axis. Subsequent investigations using resting-state fMRI and diffusion tractography in the human brain have further substantiated this specific organizational pattern (Arcaro et al. 2015 [, 2018](#)). Findings from connectivity-based parcellation studies substantially agree with this model. Anterior (and ventral) clusters predominantly exhibit connections to precentral and postcentral gyri; dorsomedial clusters show connectivity to the cingulum, precuneus and inferior parietal lobules; dorsal-lateral clusters connected to prefrontal and parietal regions and a ventromedial cluster displaying connectivity to sensorimotor and occipitotemporal cortical regions (Guedj and Vuilleumier 2020 [; Basile et al. 2024](#)).

Consistent with these findings, our study reveals that the two major axes of both structural and functional connectivity converge on the dorso-medial and ventro-lateral axes of the pulvinar complex, confirming the double dissociation pattern described therein. Ventral regions of the pulvinar (low on G_{FC1}) exhibited stronger connections to occipital and inferior temporal regions, while dorsal regions (high on G_{FC1}) were predominantly connected to frontoparietal and cingulate regions. Consequently, the secondary gradient of pulvinar-cortical functional connectivity progressed from lower extreme values in the dorsolateral pulvinar (associated with connectivity to frontoparietal regions) to dorsomedial pulvinar (mostly connected to superior temporal cortex), passing through ventromedial and ventrolateral pulvinar (respectively connected to occipital and inferior temporal lobes). Lastly, medial regions of the pulvinar (low on G_{FC3}) exhibited stronger connections to the occipital and parietal lobe compared to lateral regions (high on G_{FC3}), which were instead more correlated with superior and inferior temporal cortices.

Our findings provide a nuanced perspective on Shipp's model of pulvinar-cortical connectivity, shedding light on the intricate relationships between pulvinar-cortical and cortico-cortical connection patterns. While anatomical studies in non-human primates have suggested that highly connected cortical areas may share overlapping representations on the pulvinar complex, known as the "replication principle" (Shipp 2003 [\), our work offers robust formal evidence of this organizational principle in the human brain. Although we could not replicate the same results for structural connectivity, likely due to the inherent disproportion between pulvinar-cortical connections and the whole brain connectome, we demonstrated a one-to-one correspondence between the first three cortico-cortical functional connectivity gradients and the three main functional connectivity gradients of the pulvinar.](#)

Beyond merely confirming that cortical functional networks converge on shared connectional domains within the pulvinar complex, our findings underscore how pulvinar-cortical connectivity closely mirrors the organizational principles observed in the cerebral cortex. These principles, initially elucidated in the seminal work of Margulies and colleagues (2016) [and grounded in earlier anatomical models based on tract-tracing studies in non-human primates \(Mesulam 1998](#) [\), reflect a hierarchical patterning of information transfer from unimodal sensory or motor representation towards increasing levels of integration and abstraction in multi-modal associative cortices.](#)

Specifically, the principal gradient of cortical connectivity, extending from primary visual, somatosensory, motor, and auditory cortices to associative regions of the limbic, frontoparietal and default-mode network, not only accounted for the largest amount of variance of pulvinar functional connectivity, but also delineated the ventro-dorsal pulvinar axis. Likewise, the gradient capturing the second-most cortical connectivity variance, which spans from visual to motor-auditory unimodal cortices with modality-independent regions situated at various levels, corresponds to the third pulvinar-cortical gradient in terms of explained variance, mapped along

the medio-lateral axis. Interestingly, the third gradient of cortico-cortical connectivity, in terms of pulvinar-cortical connectivity appears slightly more represented than this secondary, cross-modal gradient, as indicated by the higher explained variance. This cortical gradient, extending from dorso-lateral to dorso-medial pulvinar regions, progresses from task-negative to task-positive networks of the cerebral cortex and has been linked to spontaneous attention during naturalistic tasks (Samara et al. 2023 [↗](#)).

Together, these results reinforce the concept of pulvinar involvement in multimodal sensory integration and top-down attentional processing, consistent with recent literature (Saalmann et al. 2012 [↗](#); Benarroch 2015 [↗](#); Fiebelkorn and Kastner 2020 [↗](#); Froesel et al. 2021 [↗](#)). They also align with the proposed role of pulvinar-cortical connectivity in enhancing communication and synchronization within cortical processing modules in a context-dependent and competitive manner (Rockland et al. 1999 [↗](#); Cortes et al. 2021 [↗](#), 2024 [↗](#); Aussel et al. 2023 [↗](#); Bastuji et al. 2024 [↗](#)).

In this context, it could be hypothesized that the observed gradient organization of the pulvinar may also exhibit specific patterns in the temporal domain. Indeed, multiple investigations have linked the temporal dynamics of cortical regions to different aspects of information processing (Rossi-Pool et al., 2021; Soltani et al., 2021). Notably, intrinsic neural timescales of functional activity have been associated with the functional specialization and gradient organization of the cerebral cortex (Golesorkhi et al., 2021), with shorter timescales in unimodal sensory regions and longer ones in transmodal networks (Ito et al., 2020; Murray et al., 2014). Moreover, thalamocortical connectivity has been showed to correlate with these patterns of intrinsic time scale (Müller et al., 2020 [↗](#)). In addition, modulatory neurotransmitters such as serotonin and dopamine have been demonstrated to play a significant role in modulating functional cortical dynamics across different timescales (Hansen, Shafiei, et al., 2022; Luppi et al., 2023). Exploring how the spatial organization of the pulvinar relates to temporal dynamics and timescale modulation could provide valuable insights and represents a promising avenue for future investigations.

In recent years, different works have explored the spatial arrangement of thalamic connectivity within a connectivity gradient framework. Diffusion embedding of thalamocortical functional connectivity has revealed a principal, medio-lateral gradient that was found correlated to thalamic structural subdivisions, and a secondary, antero-posterior gradient associated to thalamic functional subfields, and showing progression from unimodal sensorimotor cortical networks to multimodal attention and associative networks. Interestingly, the principal thalamic gradient shows a medio-lateral arrangement on the pulvinar axis while the secondary gradients corresponds more to a ventral-dorsal pulvinar axis (Yang et al. 2020 [↗](#)). In particular, further independent investigations have suggested that the progressing pattern of thalamic connectivity from unimodal to transmodal cortices is strongly associated to the local density of core and matrix cell types, thus establishing a link between molecular properties and functional connectivity dynamics (Müller et al. 2020 [↗](#); Huang et al. 2024 [↗](#)). Our findings complement and expand the existing literature by revealing a similar arrangement of cortical connectivity patterns on the pulvinar complex, and elucidating its relationship to in-vivo estimates of molecular markers of neurotransmission. We found that the gradient associated to unimodal-transmodal cortical connectivity accounted for the highest percentage of variance of variance in cortico-pulvinar connectivity, in line with its well-acknowledged role of associative nucleus. It is noteworthy that, in analyses of thalamocortical gradients, the pulvinar complex is situated towards the “sensorimotor” extreme of the unimodal-to-transmodal thalamic gradient (Yang et al., 2020 [↗](#)). This likely reflects its prominent connectivity to visual and sensory areas compared to other thalamic nuclei. Nevertheless, the extensive and intricate association of pulvinar with multiple cortical networks emerges is strongly evident in various functional connectivity investigations (Basile et al., 2021 [↗](#); Kumar et al., 2017, 2022). By isolating pulvinar-cortical from broader

thalamocortical connectivity, our analysis was able to provide additional insights into the spatial organization of its connectivity with different cortical networks, highlighting the pulvinar's remarkable functional diversity and complexity.

As regards structural connectivity, existing accounts describe a medio-lateral organization of thalamocortical connections, corresponding to an antero-posterior gradient on the cortical mantle. This gradient organization appears to be anchored to genetic markers of different cell types (Oldham and Ball 2023 [\[1\]](#)). In line with their findings, we describe a principal axis of structural connectivity in the pulvinar complex that is arranged on the mediolateral axis, and we enforce the notion of a deep relationship between structural connections and molecular expression of neurotransmission markers. On the other hand, the patterns of connectivity with the cerebral cortex do not correspond to a clear antero-posterior axis on the cerebral cortex, probably showing the predominance of local connectivity over the global thalamic structural topography. Further investigations are warranted to ascertain whether the structural gradients of the pulvinar complex may be in continuity with this general cortico-thalamic connectivity gradient.

Neurochemical correlates of pulvinar-cortical topographical organization

In addition to elucidating the intricate pulvinar-cortical connectivity patterns, our study unveils unprecedented insights into the correlates of this complex topographical organization at multiple organizational scales, including structural connectivity and neurotransmission. Notably, we found that the hierarchical organization of pulvinar-cortical connectivity aligns with the principal axis of receptor expression, indicating a substantial influence of neuromodulator receptor expression on this level of organization. This principal axis of receptor expression reflects diverging expression patterns of markers of catecholaminergic neurotransmission, encompassing dopaminergic, noradrenergic, and serotonergic systems. While previous ex-vivo and in-vivo investigations in the human and non-human primate brain confirm that pulvinar is a recipient of dopaminergic, noradrenergic, and serotonergic projections (Lavoie and Parent 1991 [\[2\]](#); Oke et al. 1997 [\[3\]](#); Rieck et al. 2004 [\[4\]](#); Sánchez-González et al. 2005 [\[5\]](#); Pérez-Santos et al. 2021 [\[6\]](#)), our work provides information on their specific distribution pattern, representing a novel contribution to the field. Moreover, studies in non-human primates have highlighted many receptor markers whose topographic distribution mirrors connectivity patterns in the pulvinar complex, including acetylcholine-esterase (AChE), parvalbumin, calbindin, or vesicular glutamate transporters 1 and 2 (vGLUT1, vGLUT2) (Gutierrez et al. 1995 [\[7\]](#), 2000 [\[8\]](#); Stepniewska and Kaas 1997 [\[9\]](#); Balaram et al. 2013 [\[10\]](#), 2015 [\[11\]](#)). Since some of these markers are known to colocalize with catecholaminergic receptors in different brain areas (Liang et al. 1996 [\[12\]](#); Graham et al. 2015 [\[13\]](#)), and have been recently correlated to functional topography in the cortex and subcortex, including the thalamus (Anderson et al. 2018 [\[14\]](#), 2020 [\[15\]](#); Müller et al. 2020 [\[16\]](#)), the macroscale pattern of receptor coexpression observed by PET may reflect broader chemoarchitectural features at the cellular level.

Indeed, while commonalities and discrepancies between structural and functional connectivity have been extensively investigated, the relationship between functional connectivity and modulatory neurotransmission remains poorly understood. Our findings reveal stronger associations between pulvinar-cortical connectivity to specific functional networks and the spatial distribution of markers of serotonergic, noradrenergic, dopaminergic and opioid systems. Pharmacological challenge studies using resting-state functional MRI suggest that each of these neurotransmission systems may either directly modulate thalamocortical connectivity (Salomon et al. 2012 [\[17\]](#); Cole et al. 2013b [\[18\]](#), a [\[19\]](#); Farr et al. 2014 [\[20\]](#); Lewis et al. 2021 [\[21\]](#); Pizzi et al. 2023 [\[22\]](#)) or influence neuronal gain in cortico-cortical functional connectivity (Shine et al. 2018 [\[23\]](#)), which is known to depend, in part, on cortical connections to associative thalamic nuclei, including the pulvinar.

Indeed, serotonergic neurotransmission in the thalamus plays a crucial role in higher-order sensory integration, especially within the visual system. In this regard, alterations in thalamo-cortical functional connectivity, influenced by powerful serotonergic psychedelics, are known to mediate this process (Preller et al. 2018 [↗](#); Gaddis et al. 2022 [↗](#); Onofri et al. 2023 [↗](#)). This perspective sheds light on the increasing 5HTT expression from high to low-level somatosensory and visual regions of the pulvinar. Conversely, the increasing expression of dopamine receptors and transporters observed in the dorsal, higher-order regions of the pulvinar is consistent with the role of dopamine neuromodulation in mediating attentional processes, as supported by the findings of increased functional connectivity between thalamus and higher-order cortical regions after administration of the selective DAT-blocker modafinil (Pizzi et al. 2023 [↗](#)). Finally, recent multi-modal investigations have reported higher levels of dopamine receptors and transporters in cortical and subcortical regions structurally and functionally connected with the dorsal attention network, including the dorsomedial pulvinar (Alves et al. 2022 [↗](#)). This further supports our results and underscores the pivotal role of dopamine neuromodulation in attentional processes.

A multimodal pulvinar gradient architecture as a possible benchmark for neurological disorders

By integrating functional and structural connectivity data with neuroreceptor expression patterns, our proposed model of multi-level gradient architecture of the pulvinar complex holds promise for elucidating alterations in pulvinar structure and connectivity observed in various pathological conditions. For instance, reduced availability of D2/3 receptors in the pulvinar correlated with functional connectivity to the superior temporal sulcus and medial occipital lobe and autistic social communication symptoms in individuals with autistic spectrum disorders (ASD) (Murayama et al. 2022 [↗](#)). Furthermore, the right anterior pulvinar (PuA), situated at the extreme of our principal coexpression gradient characterized by high expression of 5HTT and low expression of D2 receptors, exhibited reduced volume in patients with Parkinson's disease and comorbid depression. Notably, treatment with serotonergic antidepressants was associated with increased volume of this region, suggesting a protective effect of serotonergic transmission against dopamine depletion-driven degeneration (Bhome et al. 2022 [↗](#)). In another study involving first-episode psychotic patients, heightened functional connectivity to nodes of the default mode network and central executive network was observed in the dorsal pulvinar. This finding aligns with our multimodal model, wherein increased expression of D2 receptors corresponds to this region (Kwak et al. 2021 [↗](#)). We suggest that our integrative, gradient-based model of the organizational features of the pulvinar complex may provide an explorative framework to understand the role of this structure in neuropsychiatric disorders such as Lewy Body Dementia (LBD), Alzheimer's Disease, frontotemporal dementia (FTD), medial temporal lobe epilepsy (MTLE), or schizophrenia and other psychotic disorders (Guye 2006 [↗](#); Anticevic et al. 2015 [↗](#); Erskine et al. 2018 [↗](#); Capecchi et al. 2020 [↗](#); Perez-Rando et al. 2022 [↗](#); Velioglu et al. 2023 [↗](#); Zhang et al. 2023 [↗](#)), as well as the factors influencing pharmacological treatment response.

Finally, the potential to modulate activity in specific cortical functional networks based on the localization on the main pulvinar axes, as well as its relationships to structural connectivity, offers valuable insights for therapeutic interventions.

This includes deep brain stimulation of the pulvinar, an emerging area of interest in functional neurosurgery for epilepsy, which could benefit from a deeper understanding of pulvinar connectivity and its implications for treatment outcomes (Kalamatianos et al. 2023 [↗](#); Vakilna et al. 2023 [↗](#); Wong et al. 2023 [↗](#)). Specifically, the medial (as opposed to the lateral) pulvinar is the desired target due to its connectivity to the medial temporal lobe. Overall, our model opens avenues for both understanding the pathophysiology of neurological and psychiatric disorders and developing targeted therapeutic interventions.

Future perspectives and limitations

While our study offers valuable insights into the organizational principles of the pulvinar complex, it is not without limitations. One notable limitation of this study lies in the relatively small size of the pulvinar complex compared to other larger cortical or subcortical structures. The high cellular density of the pulvinar poses a challenge for the relatively coarse resolution of currently available imaging techniques. Although the generally high quality of both the main and validation datasets, including rs-fMRI data (ULurbil et al. 2013 [\[1\]](#); Babayan et al. 2019 [\[2\]](#)), align with current standards for imaging investigations of pulvinar connectivity, higher-resolution imaging approaches may offer more granular insights. Advanced techniques, such as ultra-high-field fMRI, hold promise for uncovering the fine-scale topographical organization of the pulvinar complex. The limitation of small region size is also particularly relevant for the normative PET data, that have been resampled to the voxel size of 2mm^2 to match the resolution of BOLD data and suffer from all the inherent limitations of PET imaging in small brain volumes, such as partial volume effects or tracer spillout effects (Kanel et al. 2023 [\[3\]](#)). In addition, the neurotransmitter normative atlas is derived from group-level averages of data acquired across different centers. As a consequence, the receptor density profiles employed to quantify receptor “coexpression” are measured across spatially aligned and group-averaged PET scans of different individuals (Hansen et al. 2022b [\[4\]](#)), since the radioactive nature of PET and SPECT tracers prevents the study of multiple neurotransmitter systems in the same individual for technical and clinical reasons, yet. On the other hand, this neurotransmitter atlas has been extensively validated both by correlation to an independent autoradiography dataset and to ex-vivo gene expression levels measured with microarray techniques (Hansen et al. 2022a [\[5\]](#)). As such, it therefore represents, to the best of our knowledge, the most advanced tool available to investigate the relations between neurotransmitter expression patterns in the human brain in vivo. Indeed, as mentioned above, “gold-standard” reference ex-vivo studies on molecular expression levels of the receptors and transporters analyzed in our study, relative to the pulvinar complex, are substantially sparse, and nearly absent for the human brain. Therefore, further investigations, possibly by employing immunohistochemical or immunofluorescence microscopy in the human brain coupled with highly sampled gene expression probes, are warranted to validate and enhance our understanding of molecular expression in the human pulvinar.

Conclusion

This study offers advanced insights into the interplay between pulvinar-cortical and cortico-cortical connectivity in the human brain, along with its structural and molecular correlates. Our proposed model confirms the hypothesis of a “replication principle”, illustrating that the pulvinar complex harbors multiple representations of cortico-cortical functional connectivity organized hierarchically by their information processing significance across its ventro-dorsal and medio-lateral axes. These shared representations traverse pulvinar nuclei continuously, reconciling discrete, individual nuclear connectivity and coactivation patterns with the idea of a gradient-like organization of pulvinar connectivity.

By systematically assessing the relationship between functional, structural and molecular aspects of pulvinar organization, we delineate the topographically organized features of the human pulvinar and reveal substantial convergence among the main directives of functional connectivity, anatomical connectivity and receptor coexpression. Our study underscores the pivotal role of the pulvinar complex in mediating information integration and communication across brain networks at multiple levels of brain organization. We propose that this comprehensive understanding may offer a unified explanatory framework to enhance comprehension of its involvement in higher-order perceptual and cognitive functions both in healthy patients and in various neuropsychiatric diseases.

Materials and methods

1. Data acquisition and preprocessing

1.1. Primary dataset (HCP)

Structural, diffusion-weighted, and resting-state functional MRI data of 210 healthy subjects (males=92, females=118, age range 22-36 years) were retrieved from the HCP repository (). The study protocol was approved by the Washington University in St. Louis Institutional Review Board (IRB) (Van Essen et al. 2012 [↗](#)).

178 subjects out of 210 (85%) are genetically unrelated. Of the remaining, genetically related subjects, 22 (~10% of the total sample) were included with another subject from the same family group (11 pairs); 6 (3%) were included with two other family members (2 triplets) and 4 (2%) were all parts of the same family group.

MRI data were acquired on a 3T custom-made Siemens “Connectome Skyra” scanner (Siemens, Erlangen, Germany), equipped with a Siemens SC72 gradient coil.

High-resolution T1-weighted scans (isotropic voxel size = 0.7 mm) were acquired with the following parameters: MP-RAGE sequence, TR = 2400 ms, TE = 2.14 ms (Van Essen et al. 2012 [↗](#)). Skull-stripped T1 weighted images were segmented into cortical and subcortical gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF) using FAST and FIRST FSL’s tools (Smith et al. 2004 [↗](#); Patenaude et al. 2011 [↗](#)). The MNI-space transformations available as part of the minimally preprocessed data were employed (FLIRT 12 degrees of freedom affine; FNIRT nonlinear registration) (Glasser et al. 2013a [↗](#)).

Resting-state functional MRI data (rs-fMRI) (isotropic voxel size = 2mm) were acquired with a gradient-echo EPI sequence, using the following parameters: TR = 720 ms, TE = 33.1 ms, 1200 frames, ~15 min/run. Data were acquired separately on different days along two different sessions, each session consisting of a left-to-right (LR) and a right-to-left (RL) phase encoding acquisition (Van Essen et al. 2012 [↗](#); Ulfur et al. 2013 [↗](#); Smith et al. 2013a [↗](#)). The LR and RL acquisitions of the first session only have been employed in the present work.

Data were retrieved in a minimally preprocessed form which includes artifact and motion correction, registration to 2-mm resolution MNI 152 standard space, high pass temporal filtering (> 2000 s full width at half maximum) (Glasser et al. 2013b [↗](#)), ICA-based denoising with ICA-FIX (Salimi-Khorshidi et al. 2014 [↗](#)), and regression of artifacts and motion-related parameters (Smith et al. 2013b [↗](#)). Additionally, the global WM and CSF signal was regressed out to further improve ICA-based denoising results (Plachti et al. 2019 [↗](#)).

Multi-shell DWI data (isotropic voxel size = 1.25 mm) were acquired using a single-shot 2D spin-echo multiband Echo Planar Imaging (EPI) sequence (Sotiropoulos et al. 2013 [↗](#)). The following acquisition parameters were employed: b-values: 1000, 2000, 3000 mm/s²; 90 directions per shell; spatial isotropic resolution 1.25 mm. DWI scans were retrieved in a minimally preprocessed form including eddy currents, EPI distortion and motion correction, and cross-modal linear registration of structural and DWI images (Glasser et al. 2013b [↗](#)).

A multi-shell multi-tissue (MSMT) CSD signal modeling was performed to estimate individual response functions in WM, GM, and CSF (Jeurissen et al. 2014 [↗](#)). Whole brain, probabilistic tractography (10 million streamlines, iFOD2 algorithm; anatomically constrained tractography)

was generated with MRtrix3 default parameters (Tournier et al. 2010 [↗](#); Smith et al. 2012 [↗](#)).

1.2. Validation dataset (LEMON)

Structural, diffusion, and rs-fMRI data of 213 healthy subjects (males=138, females=75, age range 20-70 years) were retrieved from the Leipzig Study for Mind-Body-Emotion Interactions (LEMON) dataset (http://fcon_1000.projects.nitrc.org/indi/retro/MPI_LEMON.html [↗](#)). The study protocol was approved by the Ethics Committee of the medical faculty of the University of Leipzig.

MRI data were acquired on a 3T scanner (MAGNETOM Verio, Siemens Healthcare GmbH, Erlangen, Germany) equipped with a 32-channel head coil was employed for MRI data acquisition. High-resolution T1-weighted (isotropic voxel size = 1 mm) scans were acquired with the following parameters: MP-RAGE sequence, TR = 5000Lms, TE = 2.92Lms. Skull-stripped T1 weighted images were segmented into cortical and subcortical GM, WM, and CSF using FAST and FIRST FSL's tools (Smith et al. 2004 [↗](#); Patenaude et al. 2011 [↗](#)). T1-weighted volumes were also non-linearly registered to the 1-mm resolution MNI 152 asymmetric template using a FLIRT 12 degrees of freedom affine transform and FNIRT non-linear registration (Jenkinson and Smith 2001 [↗](#); Jenkinson et al. 2002 [↗](#); Andersson et al. 2007 [↗](#)).

Single-shell DWI data (isotropic voxel size = 1.7 mm) were acquired using a multi-band accelerated sequence with b-value of 1000, and 60 unique diffusion-encoding directions. Data were preprocessed according to the dedicated pipeline implemented via the MRtrix3 software (Tournier et al. 2019 [↗](#)), which features denoising using Marchenko-Pastur principal component analysis (MP-PCA) (Veraart et al. 2016 [↗](#)), removal of Gibbs ringing artifacts (Kellner et al. 2016 [↗](#)); 3) eddy currents, distortion, and motion correction (Andersson et al. 2003 [↗](#); Smith et al. 2004 [↗](#); Andersson and Sotiropoulos 2016 [↗](#)) and bias field correction using the N4 algorithm (Tustison et al. 2010 [↗](#)). A single-shell 3-tissue (SS3T) CSD signal modeling was performed using MRtrix3Tissue (Dhollander et al. 2016 [↗](#)), a fork of MRtrix3 software. Whole brain, probabilistic tractography (5 million streamlines, iFOD2 algorithm; anatomically constrained tractography) was generated with default parameters (Tournier et al. 2010 [↗](#); Smith et al. 2012 [↗](#)).

For rs-fMRI data (isotropic voxel size = 2.3 mm), a gradient-echo EPI was acquired with the following parameters: TR = 1400Lms, TE = 30Lms, 15.30 min/run (Babayan et al. 2019 [↗](#)). Data were obtained in a minimally-preprocessed form, consisting of the following steps: 1) removal of the first 5 volumes to allow for signal equilibration; 2) motion and distortion correction; 3) outlier and artifact detection (rapidart) and denoising using component-based noise correction (aCompCor); 4) mean-centering and variance normalization of the time series; 5) spatial normalization to 2-mm resolution MNI 152 standard space (Babayan et al. 2019 [↗](#); Mendes et al. 2019 [↗](#)).

1.3. Positron emission tomography (PET) data

Volumetric 3D PET images were retrieved from the Network Neurosciences Lab (NetNeuroLab) github page (https://github.com/netneurolab/hansen_receptors/tree/main/data/PET_nifti_images [↗](#)). Images were collected for 19 distinct neurotransmitter receptors and transporters in a multicentric data acquisition approach resulting in multiple studies across the world. Images were acquired using optimized imaging preprocessing strategies for each radioligand (Nørgaard et al. 2019 [↗](#)). In all studies, only healthy participants were scanned, for a total of 1,238 healthy individuals (males=718, females=520). Full methodologic details about each study, the associated receptor/transporter, tracers, PET cameras, and modeling methods can be found in (Hansen et al. 2022b [↗](#)). In general, the measured estimates for each image (binding potential and/or tracer distribution volume) are proportional to receptor density and then, we will refer to them as “receptor density maps” for simplicity.

Receptor density maps, originally available at different resolutions, were resliced to the 2mm MNI152 standard template resolution and converted to z-scores. Four receptor density maps (5HT1b, D2, mGluR5, vAChT) were acquired using the same tracer in different studies and were converted to a weighted average as in the reference paper. Ten receptor/transmitters (5HT1a, 5HT1b, 5HTT, CB1, D2, DAT, GABA-A, MOR and NET) were acquired with different tracers in different studies; these receptor maps were not averaged but kept as separated, resulting in a total of 29 receptor density maps. Supplementary Material 1 summarizes details about the neurotransmitter, tracers, and number of subjects for each study.

2. Diffusion map embedding

We derived functional connectivity, structural connectivity, and receptor expression gradients for the left and right pulvinar separately. All the analyses were performed in standard space (MNI152 2006 template, 2 mm voxel size). Voxels belonging to the left or right pulvinar were defined according to the Automated Anatomical Labeling v3 Atlas (AAL3, 2mm version) (Rolls et al. 2020 [↗](#)), while cortical regions of interest (ROIs) were derived from the 400-areas version of the local-global cortical parcellation proposed by Schaefer and colleagues (Schaefer et al. 2018 [↗](#)).

Individual functional connectomes were obtained from Pearson's correlation of BOLD signal in each pulvinar voxel to the average signal within each cortical ROI. As in similar work, negative correlation values were zeroed and sparsity sampling was applied by row-wise thresholding of top 10% connectivity values, with all the values below the 90th percentile set to 0 (Margulies et al. 2016 [↗](#); Katsumi et al. 2023 [↗](#)). Functional connectivity matrices were then normalized with Fisher's r-to-z transformation and averaged across all subjects (and across LR and RL sessions, for the HCP dataset) to obtain a group-level dense functional connectome.

Structural connectomes were obtained after registration of each whole-brain tractogram to the MNI space. Streamlines connecting the pulvinar to each cortical ROI were extracted from the whole brain tractogram and mapped back to their pulvinar endpoint voxels using the *tckmap* command in MRtrix3, to obtain individual, voxel-wise pulvinar tract-density distribution maps. Such individual maps were concatenated for all cortical ROIs, converted to voxel-to-cortical ROI tract-density matrices, and averaged across all subjects to obtain a group-level dense structural connectome.

Finally, the normalized receptor density was sampled for each voxel within the pulvinar by concatenating This process resulted in a final pulvinar voxel-by-receptor density matrix, representing the expression levels of 19 neurotransmitters or receptors for each voxel within the pulvinar. This matrix will be referred to as the “coexpression matrix”.

We applied diffusion embedding (Coifman and Lafon 2006 [↗](#)), an unsupervised learning algorithm widely employed in the field of gradient mapping (Guell et al. 2018 [↗](#); Tian et al. 2020 [↗](#); Hong et al. 2020 [↗](#); Katsumi et al. 2023 [↗](#)). Following previous work, each group's asymmetric feature matrix (functional dense connectome, structural dense connectome, coexpression matrix) was converted to a symmetric voxel-by-voxel cosine distance similarity matrix (1-minus-cosine distance). For diffusion embedding, an alpha value of 0.5 was employed to maximize robustness to noise. The source code for this method is available at <https://github.com/satra/mapalign> [↗](#). Eigenvalues were utilized to quantify the variance explained by each gradient, and scree plots of explained variance against the number of gradients were employed to select the most appropriate number of gradients for each side (left or right) and modality. To address the intrinsic sign indeterminacy of gradient representations, right-sided gradients were sign-flipped to align with their left-sided counterparts if needed.

Functional and structural connectivity gradients were also obtained from cortico-cortical connectivity matrices and employed for further analysis. Functional connectivity was computed as the pairwise Pearson's correlation between the average time series of the 400 cortical ROIs. For

structural connectivity gradient embedding, the raw number of streamlines connecting each pair of cortical ROIs was extracted from the whole brain tractogram separately for the left and right hemispheres, to emphasize intra-hemispheric patterns of connectivity. Right hemisphere gradients were sign-flipped to align with their left hemisphere counterparts if necessary. The diffusion embedding analysis was performed using the same methods and parameters as for the pulvinar gradients. A graphical overview of the gradient mapping protocol is presented in [Figure 1](#).

3. Gradient analysis and statistics

To facilitate the interpretation of their significance, we performed post-hoc characterization of each pulvinar gradient at various levels.

Firstly, we assessed the involvement of discrete pulvinar nuclei in gradient organization by calculating the distribution of gradient values for each nucleus. Histological nuclei were identified using the AAL atlas, which incorporates digitalized parcellation of thalamic nuclei derived from post-mortem histology ([Iglesias et al. 2018](#)). Furthermore, to explore whether the continuous gradient architecture derived from diffusion embedding could be mapped to discrete units overlapping with pulvinar nuclei, we employed a k-means clustering strategy on gradient values, as described in previous work ([Guell et al. 2020](#)). The relevant gradients to be included in the clustering analysis were determined by identifying the elbow in their explained variance percentage graph (as in the paragraph below). For both the left and right pulvinar, the relevant gradients were then normalized within a range from 0 to 1 and concatenated. Various iterations of k-means clustering were then applied in the resulting gradient space, ranging from $k=2$ to $k=30$, and the silhouette coefficient ([Rousseeuw 1987](#)) was utilized to identify the optimal number of clusters.

Gradient clustering was performed separately for left and right pulvinar, both after concatenating relevant gradients from each single imaging modality alone (functional connectivity, structural connectivity, receptor coexpression) or by concatenating together the most relevant gradients from all the imaging modalities (combined clustering). For each single-modality and combined cluster, the similarity to anatomical pulvinar nuclei was quantified using the Dice similarity coefficient ([Dice 1945](#)).

To analyze the functional or structural connectivity gradients, we obtained gradient-weighted cortical connectivity maps by multiplying each voxel's connectivity to the 400 cortical ROIs by the corresponding gradient value. These voxel-wise, gradient-weighted pulvinar-cortical connectivity maps were then averaged to derive a single cortical projection for each pulvinar connectivity gradient.

For what concerns gradient-weighted functional connectivity maps, that are symmetric and highly similar between left and right hemisphere, left and right pulvinar-cortical connectivity maps were averaged for visualization purposes. Regarding structural connectivity maps, due to the asymmetrical distribution of pulvinar structural connectivity, that is biased towards the ipsilateral cortical hemisphere, we flipped the right gradient-weighted connectivity maps on the x axis before averaging left and right pulvinar-cortical connectivity maps, so that all the ipsilateral cortical connectivity is showed on the left side and all the contralateral connectivity is showed on the right side.

To further understand the involvement of cortical networks in each of the cortical projections of structural and functional cortical gradients, we examined the distribution of gradient-weighted connectivity values across the seven major functional cortical networks, as defined in [Thomas Yeo et al. 2011](#).

Additionally, we explored the relationship between pulvinar-cortical and cortico-cortical connectivity gradients by calculating Pearson's correlation for each gradient-weighted functional connectivity map with each cortico-cortical gradient map.

Receptor coexpression gradients were characterized by correlating voxel-wise gradient values to normalized receptor density distributions.

Finally, to investigate the relation between pulvinar gradients obtained from different modalities, we computed the pairwise Pearson's correlation between gradients obtained with different modalities.

To account for the spatial autocorrelation (SA) properties of gradient maps, for all the correlations described, statistical significance was assessed using the permutational approach described in [Burt et al. \(2020\)](#). Briefly, this method takes as input geometric distance matrices for SA estimation and involves the generation of a given number of SA-preserving permuted surrogate maps, which are then employed as nulls to estimate a permutational null distribution of the test statistic ([Burt et al. 2020](#)). Pairwise Euclidean distances between left or right pulvinar voxel coordinates were employed for pulvinar null models, while for cortical parcellated connectivity data Euclidean distances were estimated between centroids of each cortical ROI. In both cases, 1000 surrogates were generated to estimate the null distribution. Statistical tests were controlled for false discovery rate (FDR) using Benjamini and Hochberg's correction.

4. Reliability and reproducibility assessment

We conducted several analyses to assess the robustness and consistency of our main findings. Specifically, both functional and structural connectivity gradients underwent split-half and test-retest replicability analysis, as well as reproducibility analysis.

For split-half replicability analysis, the main dataset was segmented into two halves, with 105 subjects each. Subjects were randomly assigned to each half, and 100 instances of the shuffling algorithm were performed. In each iteration, individual connectivity matrices of each half were averaged into dense connectomes and underwent diffusion embedding. The average Pearson's correlation for each gradient across all the iterations was used as a reliability measure.

Test-retest replicability was based on an extension of the HCP dataset, consisting of 44 subjects (males = 13; females = 31; age range: 22–36 years) with available test-retest structural and diffusion acquisition. The dense structural and functional connectomes obtained from the test and retest datasets were compared using Pearson's correlation as a measure of similarity.

Finally, reproducibility was determined by recomputing results in the validation dataset and correlating pulvinar gradients between the primary and replication datasets.

In all the mentioned cases, gradients were realigned before correlation by using Procrustes' analysis to obtain more reliable estimates of gradient similarity ([Hong et al. 2020](#)). This approach ensured that the gradients from different datasets were aligned properly before assessing their similarity, enhancing the robustness of our analyses.

To evaluate the possible influence of SNR on connectivity-derived diffusion embeddings, we have performed a voxel-wise, modality-specific, SNR assessment to investigate correlation between spatial distribution of noise and diffusion embeddings. For each subject, we separately calculated

voxel-wise SNR maps for the left and right pulvinar, using both functional (BOLD) volumes and DWI data. For BOLD volumes, we employed the widely accepted definition of temporal signal to noise (tSNR) (Murphy et al. 2007 [↗](#))

$$tSNR = \frac{t_{mean}}{t_{std}}$$

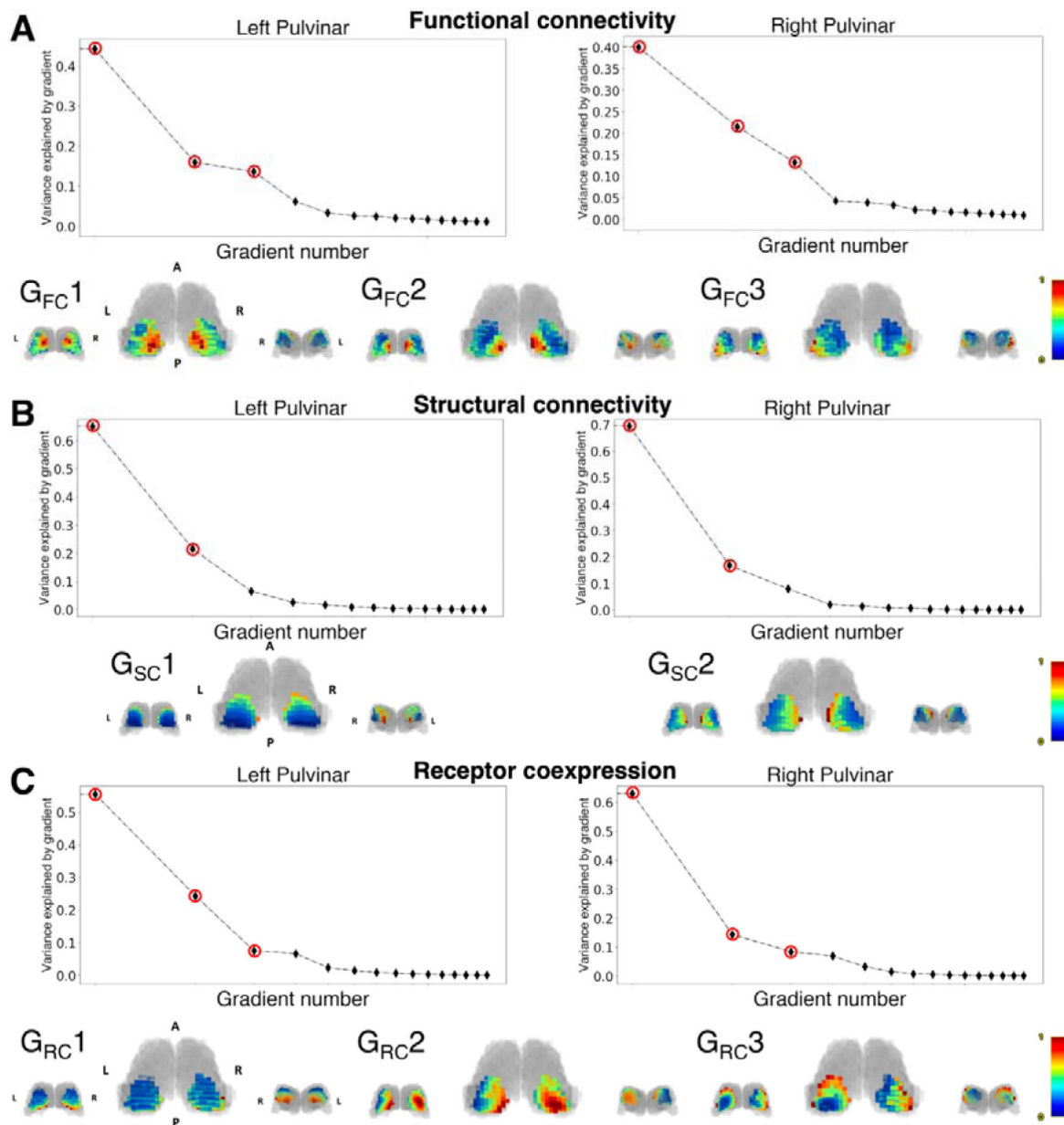
where t_{mean} and t_{std} are, respectively, the mean and the standard deviation of each voxel's signal across the time series.

For DWI data, we employed a similar definition (Cai et al. 2021 [↗](#)) that allows estimation of SNR from multiple b=0 diffusion weighted volumes:

$$SNR = \frac{S_{mean}}{S_{std}}$$

where S is the voxel's signal intensity, and the mean (S_{mean}) and standard deviation (S_{std}) were computed across all the b0-weighted volumes (18 for HCP dataset; 7 for LEMON dataset). Individual pulvinar SNR maps were then averaged to generate group-level estimates of SNR spatial distribution. The resulting, modality-specific average SNR maps were correlated with the diffusion gradients derived from the corresponding modality, following the same approach described in the previous section (Pearson's correlation; p-values corrected using spatial null models for spatial autocorrelation, and Benjamini-Hochberg correction for FWE).

Supplementary Figures



Supplementary Figure 1.

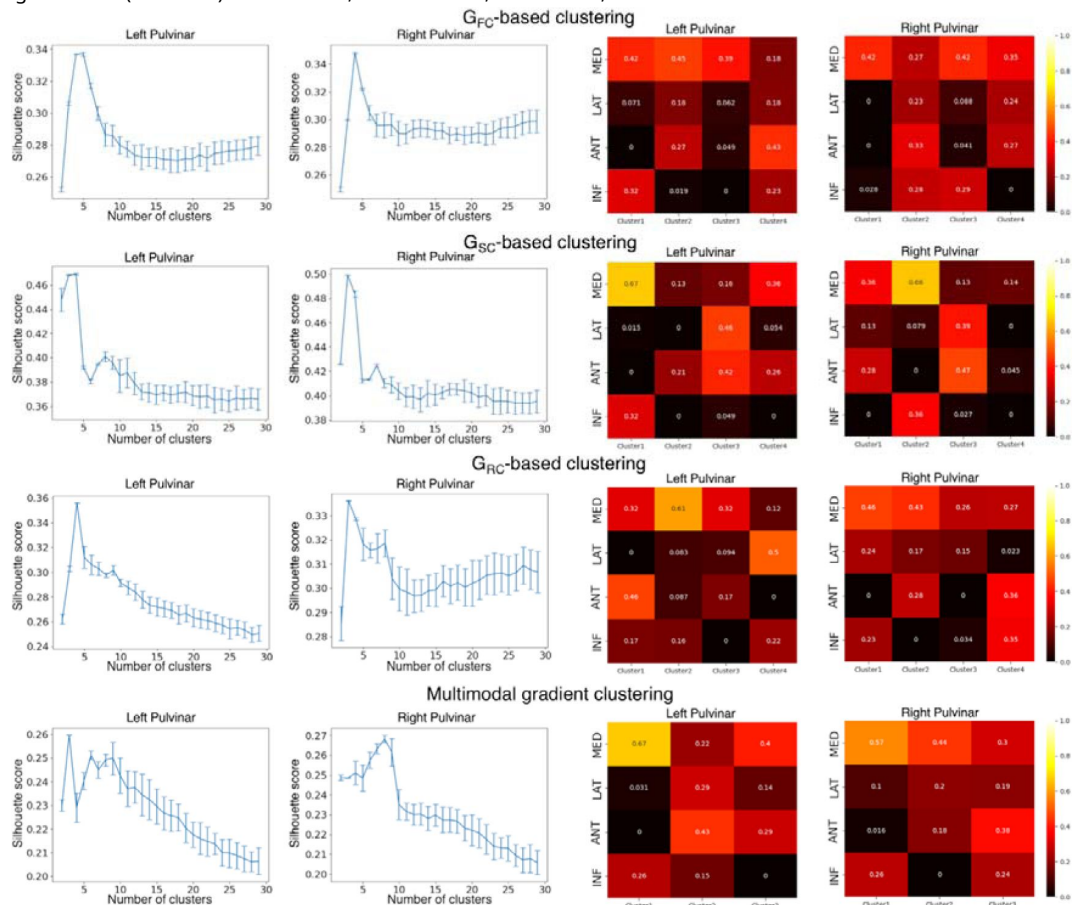
Replication of functional, structural and receptor coexpression gradients on a thalamic-specific atlas

(Su et al. 2019). For each panel, the top row shows the scree plots of explained variance for each gradient. The gradients that have been considered for following analysis are marked with red circles. The bottom row shows 3D volume renderings of the normalized gradient images overlaid on the thalamic outline (gray-shaded area). The thalamic volume has been obtained from the THOMAS atlas (Su et al. 2019). Gradients are showed in posterior (left), superior (center) and anterior (right) view. A: anterior; P: posterior; L: left; R: right.

Supplementary Figure 2.

K-means clustering of pulvinar gradients and their correspondence with histological pulvinar nuclei.

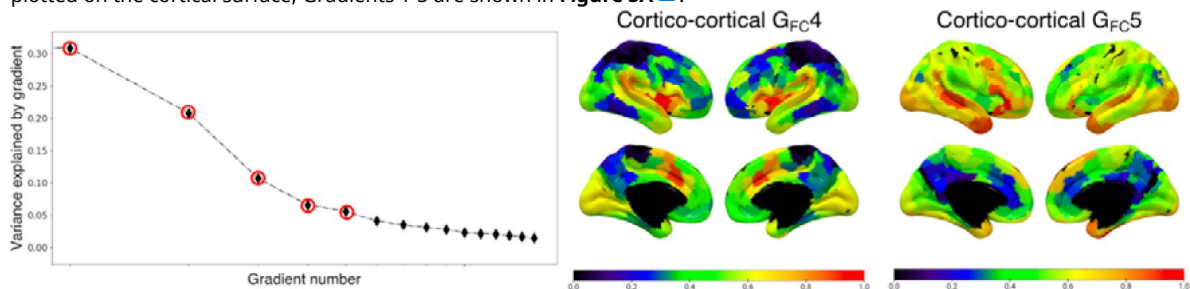
Panels to the left show the silhouette plots for left and right pulvinar clustering solutions; error bars are standard errors calculated across 50 resamples. Panels to the right show matrix plots of Dice similarity coefficients of pulvinar clusters against histological nuclei (AAL atlas). INF: inferior; ANT: anterior; LAT: lateral; MED: medial.

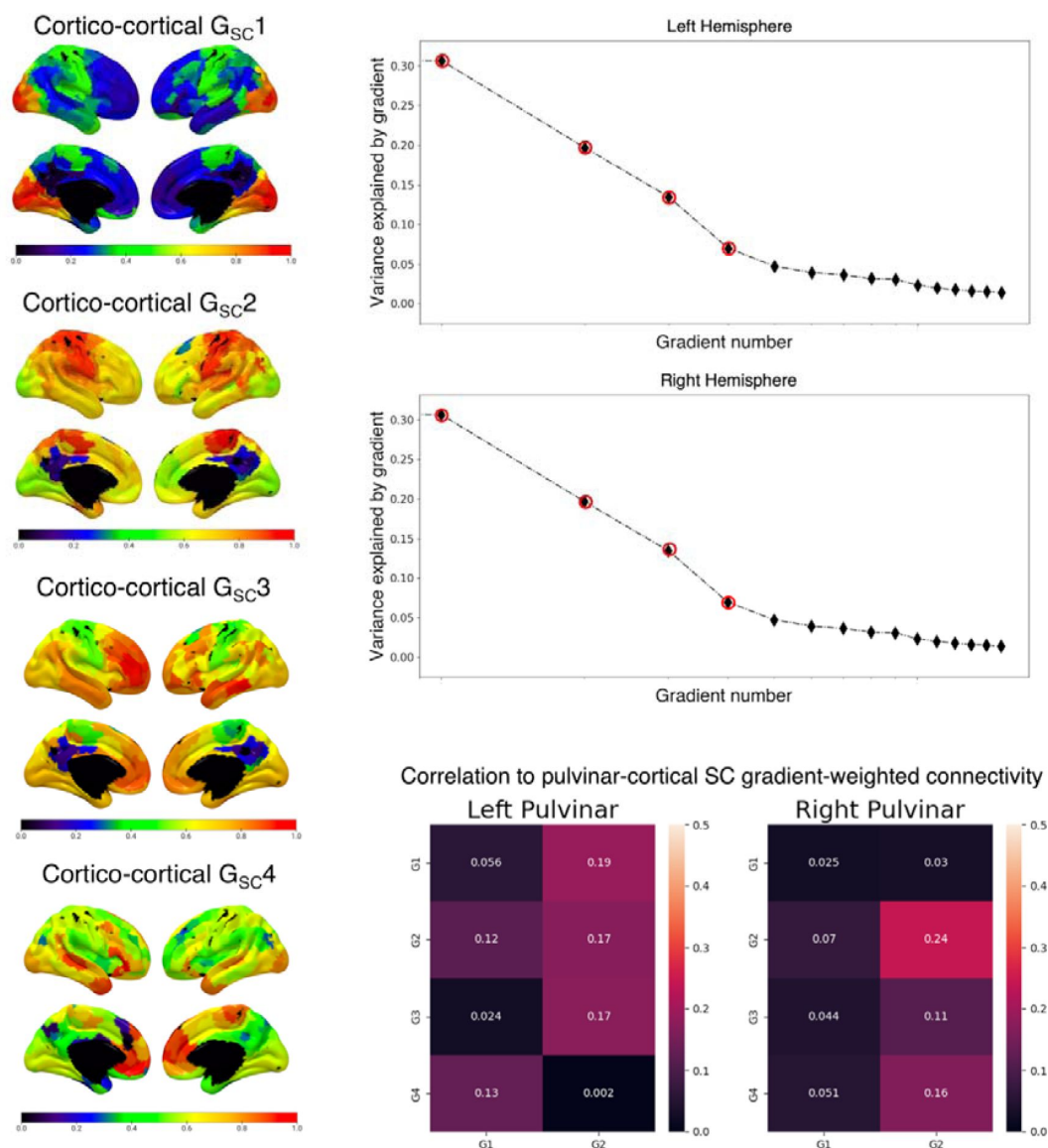


Supplementary Figure 3.

Cortico-cortical functional connectivity gradients.

The left panel shows the scree plot of explained variance for each gradient, where gradients that have been considered for subsequent analyses (1-5) are marked with red circles. The right panel shows normalized gradient values for gradients 4-5 plotted on the cortical surface; Gradients 1-3 are shown in [Figure 3A](#).

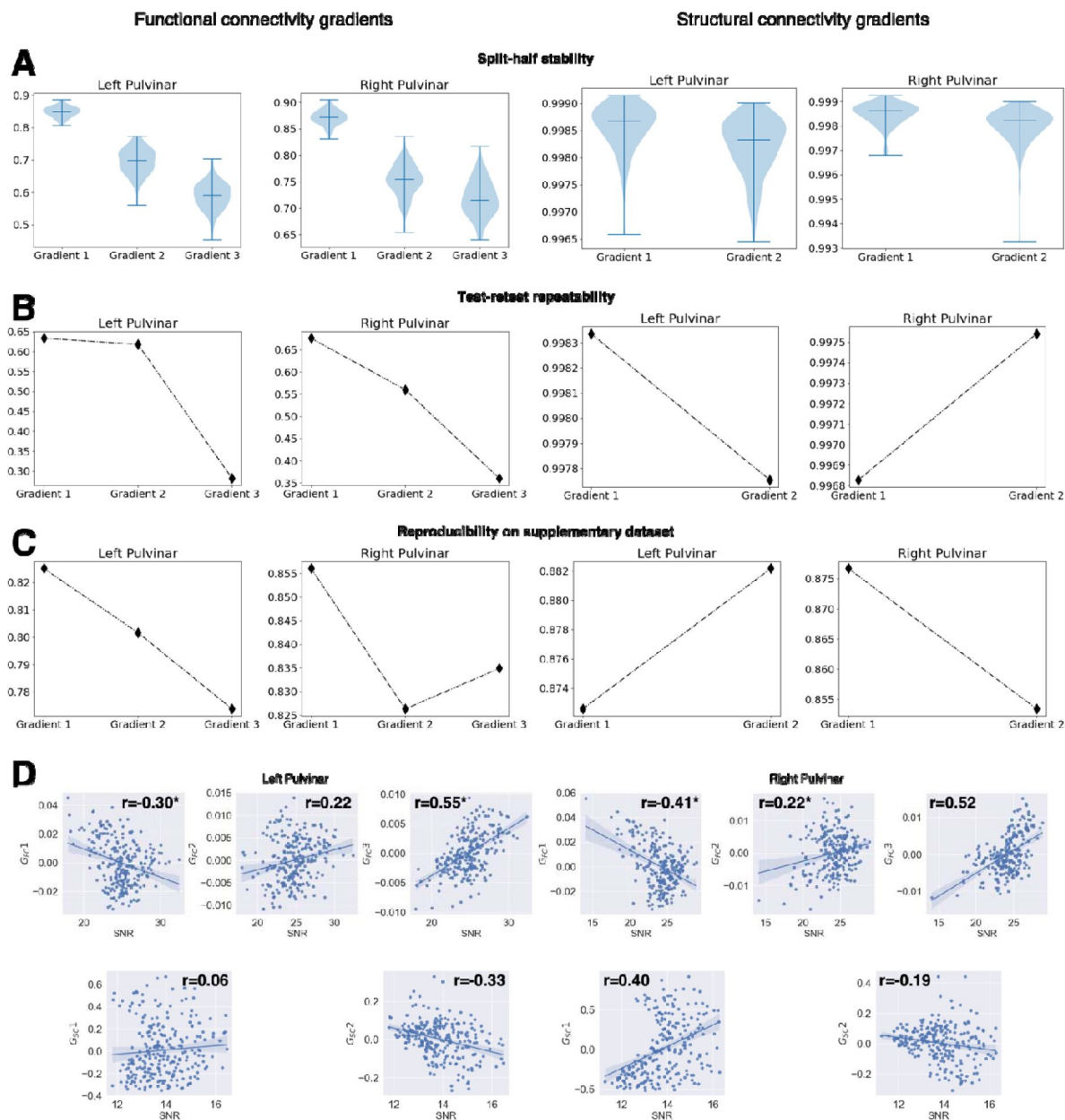




Supplementary Figure 4.

Cortico-cortical structural connectivity gradients and their relationship to pulvinar-cortical structural gradients.

The left panel shows cortico-cortical structural gradients; normalized gradient values are plotted on the cortical surface (Schaefer atlas, 400 parcels). The right panel shows scree plots of explained variance, where gradients that have been considered for subsequent analyses (1-4) are marked with red circles. Bottom right: heatmaps of Pearson's correlation values between pulvinar-cortical gradient-weighted connectivity (x-axis) and the corresponding four cortico-cortical structural connectivity gradients (y-axis).



Supplementary Figure 5.

Reliability and reproducibility of diffusion embedding of pulvinar-cortical connectivity data.

A) Split-half stability analysis. Violin plots show the distribution of correlation values across 100 split-half resamples (bars on the median and extreme values). **B) Test-retest repeatability.** **C) Reproducibility on the supplementary dataset (LEMON).**

Additional files

Supplementary Material 1 [↗](#)

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Reviewer #1 (Public review):

Summary:

The current work explored the link between the pulvinar intrinsic organisation and its functional and structural connectivity patterns of the cortex using different dimensional reduction techniques. Overall they find relationships between pulvinar-cortical organization and cortico-cortical organization, and little evidence for clustered organization. Moreover they investigate PET maps to understand how neurotransmitter/receptor distributions vary within the pulvinar and along its structural and functional connectivity axes.

Strengths:

(1) There is a replication dataset and different modalities are compared against each other to understand the structural and functional organisation of the pulvinar complex

In their revision, the authors further detailed the motivation of their study and performed various robustness checks, answering my concerns. Nevertheless, further work is needed to fully understand the role of the pulvinar nuclei and the rest of the thalamic nuclei as well as the rest of the brain, including more diverse datasets and techniques.

<https://doi.org/10.7554/eLife.100937.2.sa2>

Reviewer #2 (Public review):

Summary:

The authors aimed to explore and better understand the complex topographical organization of the human pulvinar, a brain region crucial for various high-order functions such as perception and attention. They sought to move beyond traditional histological subdivisions by investigating continuous 'gradients' of cortical connections along the dorsoventral and mediolateral axes. Using advanced imaging techniques and a comprehensive PET atlas of neurotransmitter receptors, the study aimed to identify and characterize these gradients in terms of structural connections, functional coactivation, and molecular binding patterns. Ultimately, the authors targeted to provide a more nuanced understanding of pulvinar anatomy and its implications for brain function in both healthy and diseased states.

Strengths:

A key strength of this study lies in the authors' effort to comprehensively combine multimodal data, encompassing both functional and structural connectomics, alongside the analysis of major neurotransmitter distributions. This approach enabled a more nuanced understanding of the overarching organizational principles of the pulvinar nucleus within

the broader context of whole-brain connectivity. By employing cortex-wide correlation analyses of multimodal embedding patterns derived from 'gradients,' which provide spatial maps reflecting the underlying connectomic and molecular similarities across voxels, the study offers a thorough characterization of the functional neuroanatomy of the pulvinar.

Weaknesses:

Despite its strengths, the current manuscript falls short in presenting the authors' unique perspectives on integrating the diverse biological principles derived from the various neuroimaging modalities. The findings are predominantly reported as correlations between different gradient maps, without providing the in-depth interpretations that would allow for a more comprehensive understanding of the pulvinar's role as a central hub in the brain's network.

<https://doi.org/10.7554/eLife.100937.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

The current work explored the link between the pulvinar intrinsic organisation and its functional and structural connectivity patterns of the cortex using different dimensional reduction techniques. Overall they find relationships between pulvinar-cortical organization and cortico-cortical organization, and little evidence for clustered organization. Moreover, they investigate PET maps to understand how neurotransmitter/receptor distributions vary within the pulvinar and along its structural and functional connectivity axes.

Strengths:

There is a replication dataset and different modalities are compared against each other to understand the structural and functional organisation of the pulvinar complex.

Weaknesses:

(1) What is the motivation of the study and how does this work extend previous assessments of the organization of the complete thalamus within the gradient framework?

Thank you for raising this central question. As already mentioned in the main text, pulvinar is one of the largest and prototypical associative nuclei, yet its organizational principles in the human brain remain relatively unexplored. The substantial body of anatomical research conducted in primate species suggests the coexistence of multiple coexisting and overlapping corticotopic representations on the pulvinar complex.

Existing connectivity-based parcellation studies of pulvinar organization often overlook these organizational principles, as the resulting parcellation may reflect a linear combination of single overlapping connectotopies rather than accurately capturing their distinct and unique spatial arrangement.

Investigations of thalamic connectivity have already revealed overarching organizational principles within the thalamus, which are partially reflected in its cytoarchitecture subdivision. These principles are associated with core and matrix thalamic neuronal subpopulation, and their distinct contributions to large-scale connectivity networks.

Since gradient selection relies on the explained variance of the diffusion embeddings, and pulvinar-cortical connectivity likely accounts for only a limited portion of the variance in thalamocortical connectivity, we chose to focus specifically on the pulvinar nucleus. This approach was intended to ensure that the local connectivity principles of the pulvinar are not overshadowed by the broader connectotopical organization of the entire thalamus.

This rationale aligns with findings in topographically organized regions of the cerebral cortex, such as M1, S1 or visual areas. In these regions, distinct principles of topographical organization are not readily apparent when analyzing whole-brain connectivity embedding but emerge when dimensionality reduction is applied to region-specific connectivity data.

(2) Why is the current atlas chosen for the delineation of the pulvinar and individualized maps not considered? Given the size of the pulvinar, more validation of the correctness of the atlas may be helpful.

To improve signal-to-noise ratio and in alignment with previous studies, we performed diffusion embedding on the group-level, averaged connectivity matrices rather than estimating gradients at the individual subject level.

The decision to use a standard-space atlas for pulvinar delineation, rather than individualized parcellation, was driven by technical considerations: 1) functional MRI data were already transformed to MNI space; and 2) individualized parcellation of thalamic nuclei can result in varying pulvinar volumes across subjects, complicating the averaging of connectivity data. By using a standard-space atlas, we ensured that connectivity was consistently extracted from the same set of voxels across all subjects.

We selected the AAL3 atlas (Rolls et al., 2020) over other existing thalamic atlases for practical reasons: the atlas incorporates an ex-vivo thalamic parcellation (Iglesias et al., 2018) with a specific delineation of pulvinar nuclei, which was necessary for subsequent analyses. In the revised version of the manuscript, to validate our findings, we replicated the pulvinar gradient using a different pulvinar delineation from a recent, thalamus-specific atlas (Su et al., 2019). Notably, the spatial distribution of pulvinar connectivity and coexpression gradients remained consistent, regardless of the choice of the thalamic atlas, underscoring the robustness of our results.

(3) Overall the study feels a little incremental and a repetition of what others have done already in the thalamus. It would be good to know how focusing only on the pulvinar changes interpretation, for example by comparing thalamic and pulvinar gradients?

The authors acknowledge the existing body of literature that has examined thalamic connectivity under the lens of the connectivity gradient framework. While these studies may provide valuable insights into the functional topography of the pulvinar complex -given its prominent role within the thalamus - we contend that a focused analysis of pulvinar connectivity offers a unique opportunity to uncover the specific organization principles of this nuclear complex. By isolating the pulvinar, we aimed to avoid the potential overshadowing of its local connectivity patterns by the broader connectotopical organization of the entire thalamus. However, as we believe that our findings are best interpreted within the broader context of general thalamic connectivity organization, we have included an additional paragraph in the Discussion section, which explores the similarities and

differences between thalamic and pulvinar gradients, offering a more integrative perspective on our results.

“In recent years, different works have explored the spatial arrangement of thalamic connectivity within a connectivity gradient framework. Diffusion embedding of thalamocortical functional connectivity has revealed a principal, medio-lateral gradient that was found correlated to thalamic structural subdivisions, and a secondary, antero-posterior gradient associated with thalamic functional subfields, and showing progression from unimodal sensorimotor cortical networks to multimodal attention and associative networks. Interestingly, the principal thalamic gradient shows a medio-lateral arrangement on the pulvinar axis while the secondary gradients correspond more to a ventral-dorsal pulvinar axis (Yang et al. 2020). In particular, further independent investigations have suggested that the progressing pattern of thalamic connectivity from unimodal to transmodal cortices is strongly associated to the local density of core and matrix cell types, thus establishing a link between molecular properties and functional connectivity dynamics (Müller et al. 2020; Huang et al. 2024). Our findings complement and expand the existing literature by revealing a similar arrangement of cortical connectivity patterns on the pulvinar complex, and elucidating its relationship to in-vivo estimates of molecular markers of neurotransmission. We found that the gradient associated to unimodal-transmodal cortical connectivity accounted for the highest percentage of variance of variance in cortico-pulvinar connectivity, in line with its well-acknowledged role of associative nucleus. It is noteworthy that, in analyses of thalamocortical gradients, the pulvinar complex is situated towards the “sensorimotor” extreme of the unimodal-to-transmodal thalamic gradient (Yang et al., 2020). This likely reflects its prominent connectivity to visual and sensory areas compared to other thalamic nuclei. Nevertheless, the extensive and intricate association of pulvinar with multiple cortical networks emerges is strongly evident in various functional connectivity investigations (Basile et al., 2021; Kumar et al., 2017, 2022). By isolating pulvinar-cortical from broader thalamocortical connectivity, our analysis was able to provide additional insights into the spatial organization of its connectivity with different cortical networks, highlighting the pulvinar’s remarkable functional diversity and complexity.”

(4) *Could it be that the gradient patterns stem from lacking anatomical and functional resolutions (or low SNR) therefore generating no sharp boundaries?*

The gradient organization described in our results is aligns with anatomical evidence on non-human primates (Shipp, 2003), and with existing neuroimaging studies in humans, which report limited correspondence between connectivity-based hard clustering solutions and histological delineation of pulvinar nuclei. However, we recognize the critical importance of assessing the impact of SNR on connectivity measures derived from functional and structural MRI. In the revised manuscript, we have included an additional analysis to investigate the potential impact of local noise on gradient reconstruction. This analysis involved sampling voxel-wise SNR estimates in the pulvinar from both BOLD and diffusion-weighted MRI data, averaging these estimates to generate group-level, modality-specific SNR maps. We then assessed spatial correlations between these maps and the gradient embeddings using the same methodological framework employed throughout the study. Our findings indicate that functional connectivity gradients are weakly, but significantly correlated to SNR, with the strongest correlation observed for the third gradient (left hemisphere G_{FC1} $r = -0.30$, SA-corrected $p < 0.001$, G_{FC2} $r = 0.22$, SA-corrected $p = 0.05$, G_{FC3} $r = 0.55$, SA-corrected $p < 0.001$; right hemisphere G_{FC1} $r = -0.41$, SA-corrected $p < 0.001$, G_{FC2} $r = 0.22$, SA-corrected $p = 0.008$, G_{FC3} $r = 0.52$, SA-corrected $p = 0.017$). In contrast, structural connectivity gradients showed no significant correlation with SNR (left hemisphere G_{SC1} $r = 0.06$, SA-corrected $p = 0.82$, G_{SC2} $r = -0.33$, SA-corrected $p = 0.01$; right hemisphere G_{SC1} $r = 0.40$, SA-corrected $p = 0.28$, G_{SC2} $r = -0.19$, SA-corrected $p = 0.31$).

Reviewer #1 (Recommendations for the authors):

(1) Please add more literature on thalamus gradients and interpret this with care.

Thank you for the suggestion. We have added the following paragraph in the Discussion section:

“In recent years, different works have explored the spatial arrangement of thalamic connectivity within a connectivity gradient framework. Diffusion embedding of thalamocortical functional connectivity has revealed a principal, medio-lateral gradient that was found correlated to thalamic structural subdivisions, and a secondary, antero-posterior gradient associated with thalamic functional subfields, and showing progression from unimodal sensorimotor cortical networks to multimodal attention and associative networks. Interestingly, the principal thalamic gradient shows a medio-lateral arrangement on the pulvinar axis while the secondary gradients correspond more to a ventral-dorsal pulvinar axis (Yang et al. 2020). In particular, further independent investigations have suggested that the progressing pattern of thalamic connectivity from unimodal to transmodal cortices is strongly associated to the local density of core and matrix cell types, thus establishing a link between molecular properties and functional connectivity dynamics (Müller et al. 2020; Huang et al. 2024). Our findings complement and expand the existing literature by revealing a similar arrangement of cortical connectivity patterns on the pulvinar complex, and elucidating its relationship to in-vivo estimates of molecular markers of neurotransmission. We found that the gradient associated to unimodal-transmodal cortical connectivity accounted for the highest percentage of variance of variance in cortico-pulvinar connectivity, in line with its well-acknowledged role of associative nucleus. It is noteworthy that, in analyses of thalamocortical gradients, the pulvinar complex is situated towards the “sensorimotor” extreme of the unimodal-to-transmodal thalamic gradient (Yang et al., 2020). This likely reflects its prominent connectivity to visual and sensory areas compared to other thalamic nuclei. Nevertheless, the extensive and intricate association of pulvinar with multiple cortical networks emerges is strongly evident in various functional connectivity investigations (Basile et al., 2021; Kumar et al., 2017, 2022). By isolating pulvinar-cortical from broader thalamocortical connectivity, our analysis was able to provide additional insights into the spatial organization of its connectivity with different cortical networks, highlighting the pulvinar’s remarkable functional diversity and complexity.

As regards structural connectivity, existing accounts describe a medio-lateral organization of thalamocortical connections, corresponding to an antero-posterior gradient on the cortical mantle. This gradient organization appears to be anchored to genetic markers of different cell types (Oldham and Ball 2023). In line with their findings, we describe a principal axis of structural connectivity in the pulvinar complex that is arranged on the mediolateral axis, and we enforce the notion of a deep relationship between structural connections and molecular expression of neurotransmission markers. On the other hand, the patterns of connectivity with the cerebral cortex do not correspond to a clear antero-posterior axis on the cerebral cortex, probably showing the predominance of local connectivity over the global thalamic structural topography. Further investigations are warranted to ascertain whether the structural gradients of the pulvinar complex may be in continuity with this general cortico-thalamic connectivity gradient.”

(2) Please state the motivation of the work more clearly and what makes it different from related literature.

Thank you for pointing us to this lack of clarity. We have added the following paragraph in the Introduction section:

“In particular, investigations of thalamic connectivity within the gradient framework have uncovered general organizational principles within the thalamus, which are partially reflected in thalamic cytoarchitecture subdivisions. These principles have been linked to core and matrix thalamic neuronal subpopulation, and to their differential contribution to large-scale connectivity networks (Müller et al., 2020; Yang et al., 2020). However, given the remarkable functional complexity and diversity of the pulvinar complex, these global spatial organization patterns likely capture only part of its functional topography. With this in mind, isolating pulvinar connectivity from the remaining thalamocortical connectome would ensure that local organizational principles are not obscured by the global connectotopic structure of the entire thalamus.”

(3) Why did the authors opt for a whole brain labelling atlas, would a thalamus-specific atlas not be more suitable?

Despite being a large-scale whole brain atlas, the labeling atlas of choice (AAL3) incorporates a thalamus-specific parcellation from previous work (Iglesias et al., 2018), derived from ex-vivo data and including subdivision of the pulvinar complex into anterior, inferior, lateral and medial nuclei. In the revised version of the manuscript, to validate our findings, we replicated the pulvinar gradient using a different pulvinar delineation from a recent, thalamus-specific atlas (Su et al., 2019). We show these results in Supplementary Figure 1. Notably, the spatial distribution of pulvinar connectivity and coexpression gradients remained consistent, regardless of the choice of the thalamic atlas, underscoring the robustness of our results.

(4) How did the authors account for the potential low sensitivity of subcortical signals in the PET data?

We acknowledge the inherent limitations in spatial sensitivity that are a common drawback of PET imaging. However, the PET data employed in the present study were derived from a high-quality dataset collected across multiple studies, predominantly acquired using high resolution scanners (Hansen et al., 2022; see supplementary material at https://static-content.springer.com/esm/art%3A10.1038%2Fs41593-022-01186-3/MediaObjects/41593_2022_1186_MOESM3_ESM.xlsx for technical details). Furthermore, the reliability of neurotransmission markers measurements at the subcortical level has been validated against genetic transcription markers (Hansen, Markello, et al., 2022; Hansen, Shafiei, et al., 2022), ensuring robust and biologically meaningful results.

(5) What about SNR of the metrics within the pulvinar?

The referee raises a crucial and complex point, prompting us to conduct additional analyses. We recognize the critical importance of assessing the impact of SNR on connectivity measures derived from functional and structural MRI. In the revised manuscript, we have included an additional analysis to investigate the potential impact of local noise on gradient reconstruction. Therefore, we have incorporated the following text into the manuscript:

Results (5. Reliability and Reproducibility):

“To assess the influence of local noise on functional and structural connectivity gradients, we calculated the spatial correlation between gradient values and averaged voxel-wise estimates of signal-to-noise ratio (SNR) from functional and structural MRI data, respectively. We found that functional connectivity gradients are weakly, but significantly correlated with the SNR, with the strongest correlation observed for the third gradient (left hemisphere G_{FC1} $r = -0.30$, SA-corrected $p < 0.001$, G_{FC2} $r = 0.22$, SA-corrected $p = 0.05$, G_{FC3} $r = 0.55$, SA-corrected $p < 0.001$; right hemisphere G_{FC1} $r = -0.41$, SA-corrected $p < 0.001$, G_{FC2} $r = 0.22$, SA-corrected $p =$

0.008, G_{FC3} $r = 0.52$, SA-corrected $p = 0.017$). In contrast, structural connectivity gradients were not significantly associated with SNR (left hemisphere G_{SC1} $r = 0.06$, SA-corrected $p = 0.82$, G_{SC2} $r = -0.33$, SA-corrected $p = 0.01$; right hemisphere G_{SC1} $r = 0.40$, SA-corrected $p = 0.28$, G_{SC2} $r = -0.19$, SA-corrected $p = 0.31$) (Supplementary Figure 5)."

Methods (4. Reliability and reproducibility assessment):

"To evaluate the possible influence of SNR on connectivity-derived diffusion embeddings, we have performed a voxel-wise,

modality-specific, SNR assessment to investigate correlation between spatial distribution of noise and diffusion embeddings. For each subject, we separately calculated voxel-wise SNR maps for the left and right pulvinar, using both functional (BOLD) volumes and DWI data. For BOLD volumes, we employed the widely accepted definition of temporal signal to noise (tSNR) (Murphy et al., 2006):

$$tSNR = \frac{\mu_{mean}}{\sigma_{std}}$$

where T_{mean} and T_{std} are, respectively, the mean and the standard deviation of each voxel's signal across the time series.

For the DWI data, we applied a similar approach (Cai et al., 2021) that allows estimation of SNR from multiple $b=0$ diffusion weighted volumes:

$$SNR = \frac{S_{mean}}{S_{std}}$$

where S is the voxel's signal intensity, and the mean (S_{mean}) and standard deviation (S_{std}) were computed across all the b_0 -weighted volumes (18 for HCP dataset; 7 for LEMON dataset). Individual pulvinar SNR maps were then averaged to generate group-level estimates of SNR spatial distribution. The resulting, modality-specific average SNR maps were correlated with the diffusion gradients derived from the corresponding modality, following the same approach described in the previous section (Pearson's correlation; p -values corrected using spatial null models for spatial autocorrelation, and Benjamini-Hochberg correction for FWE)."

(6) The numbers of the screeplot / numbers in figures are quite small and not so easy to read.

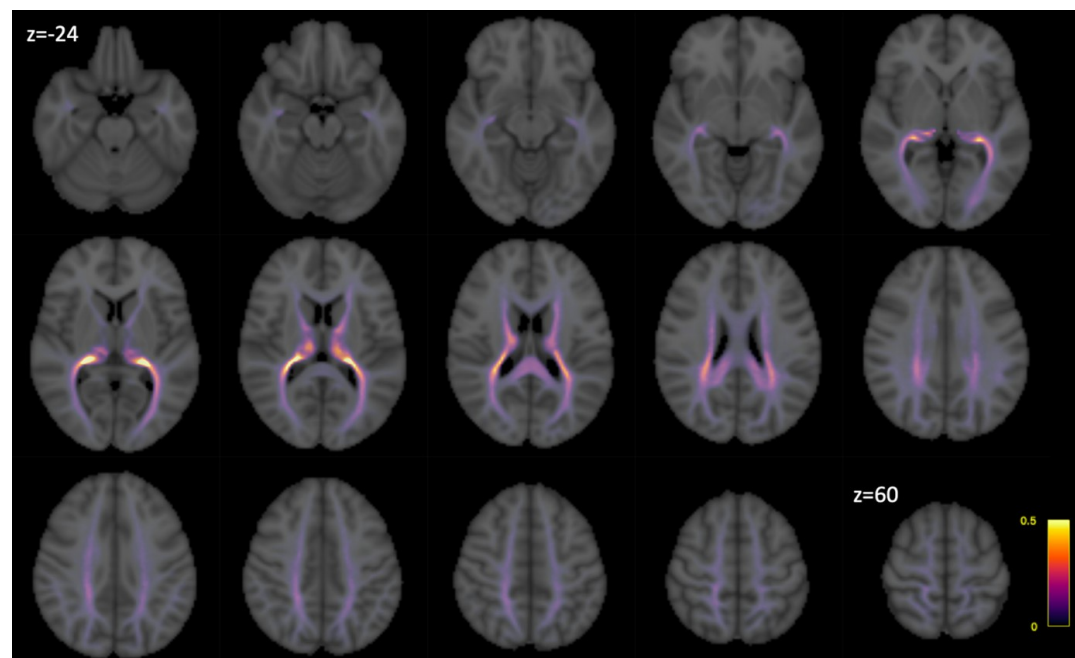
Thank you for highlighting this point. We have fixed this issue in the revised version of the Figures.

(7) How do you know the pulvinar mask is not also picking up on the cortical spinal tract?

To ensure that pulvinar masks did not pick up streamlines from the corticospinal tracts, we performed a thorough visual inspection of the tractograms that were employed for structural connectivity estimation. For each subject-specific tractogram, we randomly subsampled 10000 streamlines after transformation into MNI standard space and summed up these results to generate a group-level tractogram in standard space. The resulting track-density images (Author response image 1) demonstrate only minimal involvement of descending/ascending tracts from/to the brainstem and spinal cord, confirming the specificity of the pulvinar masks.

Author response image 1.

Group-level structural connectivity of the pulvinar complex. Track-density images have been normalized and overlaid on the MNI152 standard template.



(8) There is no mention of the within pulvinar gradients that then are correlated with PET patterns or across gradients are tested to spatial autocorrelation? I believe it is only mentioned for the cortex.

Thanks for providing us with the opportunity to clarify this important aspect, which is mentioned in the Methods section (3. Gradient analysis and statistics):

“To account for the spatial autocorrelation (SA) properties of gradient maps, for all the correlations described, statistical significance was assessed using the permutational approach described in Burt et al. (2020). Briefly, this method takes as input geometric distance matrices for SA estimation and involves the generation of a given number of SA-preserving permuted surrogate maps, which are then employed as nulls to estimate a permutational null distribution of the test statistic (Burt et al. 2020). Pairwise Euclidean distances between left or right pulvinar voxel coordinates were employed for pulvinar null models, while for cortical parcellated connectivity data Euclidean distances were estimated between centroids of each cortical ROI. In both cases, 1000 surrogates were generated to estimate the null distribution. Statistical tests were controlled for false discovery rate (FDR) using Benjamini and Hochberg’s correction.”

However, to enhance readability, we have highlighted this concept in the Results section (3. The unimodal-to-transmodal gradient (G_{FC1}) aligns with receptor expression on the dorso-ventral pulvinar axis):

“To take into account the effects of spatial autocorrelation, we corrected the resulting p-values using a method based on SA-preserving spatial null models (Burt et al. 2020)”.

(9) I don't fully understand why the mappings are so patchy of the structural connectivity gradient? Maybe some normalisation went wrong? Other papers on thalamic gradients

show smoother patterns.

We thank the Reviewer for the observation. After thoroughly reviewing the related codes, we found no normalization errors. However, we identified a visualization issue, which has been addressed in the revised version. Specifically, the structural gradient representations showed in the figures were based on the averaged values of left and right pulvinar gradients both of which include structural connectivity to either the ipsilateral or contralateral cerebral cortex. Since ipsilateral connectivity is more prominently represented than contralateral connectivity, this led to asymmetric gradient patterns between ipsilateral and contralateral cortical gradients, resulting in a patchy representation when gradients were averaged between left and right pulvinar. To resolve this, we adjusted the visualization by flipping the right pulvinar gradient representations along the x axis, aligning all the ipsilateral cortical connectivity on the left side and all the contralateral connectivity on the right. This adjustment produced smoother, more readable, and interpretable visualizations. Additionally, it allowed the asymmetry between ipsilateral and contralateral connections to be more clearly appreciated.

(10) The final statement of the abstract is misleading as we at this point don't know how making spatial pattern maps in the pulvinar may help understand the role of the pulvinar in health and disease.

We appreciate the Reviewer's suggestion and have updated the expression accordingly:

"Our findings represent a significant step forward in advancing the understanding of pulvinar anatomy and function, offering an exploratory framework to investigate the role of this structure in both health and disease."

Reviewer #2 (Public review):

Summary:

The authors aimed to explore and better understand the complex topographical organization of the human pulvinar, a brain region crucial for various high-order functions such as perception and attention. They sought to move beyond traditional histological subdivisions by investigating continuous 'gradients' of cortical connections along the dorsoventral and mediolateral axes. Using advanced imaging techniques and a comprehensive PET atlas of neurotransmitter receptors, the study aimed to identify and characterize these gradients in terms of structural connections, functional coactivation, and molecular binding patterns. Ultimately, the authors targeted to provide a more nuanced understanding of pulvinar anatomy and its implications for brain function in both healthy and diseased states.

Strengths:

A key strength of this study lies in the authors' effort to comprehensively combine multimodal data, encompassing both functional and structural connectomics, alongside the analysis of major neurotransmitter distributions. This approach enabled a more nuanced understanding of the overarching organizational principles of the pulvinar nucleus within the broader context of whole-brain connectivity. By employing cortex-wide correlation analyses of multimodal embedding patterns derived from 'gradients,' which provide spatial maps reflecting the underlying connectomic and molecular similarities across voxels, the study offers a thorough characterization of the functional neuroanatomy of the pulvinar.

Weaknesses:

Despite its strengths, the current manuscript falls short in presenting the authors' unique perspectives on integrating the diverse biological principles derived from the various neuroimaging modalities. The findings are predominantly reported as correlations between different gradient maps, without providing the in-depth interpretations that would allow for a more comprehensive understanding of the pulvinar's role as a central hub in the brain's network. Another limitation of the study is the lack of clarity regarding the application of pulvinar and its subnuclei segmentation maps to individual brains prior to BOLD signal extraction and gradient reconstruction. This omission raises concerns about the precision and reproducibility of the findings, leaving their robustness less transparently evaluable.

We thank the Reviewer for the valuable comments. While commonalities and discrepancies between structural and functional connectivity have been extensively explored in the literature, the relationship between functional connectivity and modulatory neurotransmission remains poorly understood. Specifically, while the role of thalamic modulatory neurotransmission has been thoroughly investigated in experimental animal models from an electrophysiological perspective, it remains relatively underexplored in the human brain. In our study, we identified significant associations between the spatial distribution of serotonergic, noradrenergic, dopaminergic and mu-opioid systems and functional pulvinar-cortical connectivity to specific functional networks. Evidence from pharmacological challenge studies using resting-state fMRI suggests that these neurotransmission systems may modulate network-specific thalamocortical connectivity directly or influence neural gain in cortico-cortical connectivity, a process partially dependent on thalamocortical connections to associative thalamic nuclei. However, the limitations of spatial and receptor specificity inherent to this approach, coupled with the predominantly correlational nature of our study design, prevented us from drawing more definitive conclusions on the biological relationship between neurotransmitter expression and functional connectivity. As regards the lack of clarity concerning signal extraction, we have now clarified that all the relevant steps of time series extraction were performed in standard space, without any further registration to individual subjects.

Reviewer #2 (Recommendations for the authors):

In line with the weaknesses that I raised above, my recommendation to authors are two-fold:

(1) Please provide readers with a more holistic viewpoint to better digest all the correlation analyses. For instance, in p18, the summary says:

"G_{FC}1, GRC1, and G_{SC}2 substantially delineate multiscale differences between the ventral and dorsal aspects of the pulvinar. Moving along the ventral-dorsal axis of the pulvinar complex, more ventral regions showed higher functional connectivity to unimodal sensory processing networks, higher levels of 5HTT and NAT expression, and preferentially higher structural connectivity to modality-independent or low-level sensory processing cortices."

We already knew somehow the existence of the dorsoventral axis in the pulvinar, as the authors already specified in the introduction. Beyond this simple report on phenomenological observation, one may provide a more integrated discussion to pinpoint what commonality or discrepancy the GFC, GRC, and GSC map show and potential common principles explaining their biological relationship (e.g., the 5HTT and NAT's high expression and functional connectivity). Such digested perspectives will grant the study unique insights into the functional system of the pulvinar.

We have expanded on this topic in the Discussion section (Neurochemical correlates of pulvinar-cortical topographical organization) as follows:

“Indeed, while commonalities and discrepancies between structural and functional connectivity have been extensively investigated, the relationship between functional connectivity and modulatory neurotransmission remains poorly understood. Our findings reveal stronger associations between pulvinar-cortical connectivity to specific functional networks and the spatial distribution of markers of serotonergic, noradrenergic, dopaminergic and opioid systems. Pharmacological challenge studies using resting-state functional MRI suggest that each of these neurotransmission systems may either directly modulate thalamocortical connectivity or influence neuronal gain in cortico-cortical functional connectivity, which is known to depend, in part, on cortical connections to associative thalamic nuclei, including the pulvinar.”

(2) *Specify the details if there was a QC procedure to check the signal extraction from the pulvinar subnuclei by applying the segmentation atlas at each individual.*

Preprocessed BOLD volumes were available in standard-space, and time series were extracted for each voxel within a standard-space mask of the pulvinar complex. All volumes underwent visual inspection to ensure the accuracy of the registration process. Regarding the pulvinar subnuclei, these structures were not segmented at the individual level.

Reviewer #3 (Public review):

Summary of the Study:

The authors investigate the organization of the human pulvinar by analyzing DWI, fMRI, and PET data. The authors explore the hypothesis of the "replication principle" in the pulvinar.

Strengths and Weaknesses of the Methods and Results:

The study effectively integrates diverse imaging modalities to provide a view of the pulvinar's organization. The use of analysis techniques, such as diffusion embedding-driven gradients combined with detailed interpretations of the pulvinar, is a strength.

Even though the study uses the best publicly available resolution possible with current MR-technology, the pulvinar is densely packed with many cell bodies, requiring even higher spatial resolution. In addition, the model order selection of gradients may vary with the acquired data quality. Therefore, the pulvinar's intricate organization needs further exploration with even higher spatial resolution to capture gradients closer to the biological organization of the pulvinar.

Appraisal of the Study's Aims and Conclusions:

The authors delineate the gradient organization of the pulvinar. The study provides a basis for understanding the pulvinar's role in mediating brain network communication.

Impact and Utility of the Work:

This work contributes to the field by offering insights into pulvinar organization.

We thank the Reviewer for their positive assessment and constructive feedback. The Authors agree with the Reviewer that the spatial resolution of currently available in-vivo imaging methods is limited, and that gradient representation would indeed benefit from higher resolution data. However, we also note that the resolution of structural and functional

volumes used in our study is consistent with existing literature on pulvinar connectivity. Additionally, the PET data employed in our work include multi-centric studies collected worldwide from healthy populations, and are primarily acquired using high-resolution scanners that allow spatial resolution up to 2 mm². Notwithstanding, further investigations employing finer resolution imaging techniques, such as ultra-high field fMRI, may provide more detailed insights into pulvinar topographical organization at a finer scale.

Reviewer #3 (Recommendations for the authors):

(1) The HCP data contains genetically related datasets. Please mention whether the data-selection criteria for the selected 210 healthy subjects followed the genetically unrelated criteria.

The HCP sample employed in this study consists of an initial cohort of 100 unrelated subjects, as provided in the HCP database, along with an additional random sample of 110 subjects. Subjects were selected without following a genetic criterion, as the family structure of the HCP dataset was part of a restricted access subset that we did not have access to at the time of processing. Subsequently, we obtained access to this information and determined that 178 out of 210 subjects (85%) are genetically unrelated. Of the remaining, genetically related subjects, 22 (~10% of the total sample) were included with another subject from the same family group (11 pairs); 6 (3%) were included with two other family members (2 triplets) and 4 (2%) were all parts of the same family group. This information has been included in the Methods section for clarity.

(2) The study uses HCP data with an fMRI resolution of 2mm isotropic and diffusion MRI with 1.25mm. Additionally, the LEMON dataset includes 1.7mm isotropic DWI data and fMRI with 2.3mm isotropic resolution. Furthermore, the available PET data from the Hansen et al. 2022b study has a rather coarser spatial resolution. Therefore, it may be important to mention in the discussion that the pulvinar is densely packed with cell bodies and that their gradient organization might be better reflected with even higher spatial resolution or improved measurement techniques used in the study.

We have revised the conclusive section of the Discussion into a paragraph title “Future perspectives and limitations”, and added the following text:

“One notable limitation of this study lies in the relatively small size of the pulvinar complex compared to other larger cortical or subcortical structures. The high cellular density of the pulvinar poses a challenge for the relatively coarse resolution of currently available imaging techniques. Although the generally high quality of both the main and validation datasets, including rs-fMRI data (Uğurbil et al. 2013; Babayan et al. 2019), align with current standards for imaging investigations of pulvinar connectivity, higher-resolution imaging approaches may offer more granular insights. Advanced techniques, such as ultra-high-field fMRI, hold promise for uncovering the fine-scale topographical organization of the pulvinar complex.”

(3) The functional multiplicity of the Pulvinar nuclei among other thalamus nuclei is also illustrated in <https://doi.org/10.1038/s42003-022-04126-w>

We thank the Reviewer for suggesting this important reference. We have added the following text in the Discussion section:

“It is noteworthy that, in analyses of thalamocortical gradients, the pulvinar complex is situated towards the “sensorimotor” extreme of the unimodal-to-transmodal thalamic gradient (Yang et al., 2020). This likely reflects its prominent connectivity to visual and sensory areas compared to other thalamic nuclei. Nevertheless, the extensive and intricate association of pulvinar with multiple cortical networks emerges is strongly evident in various

functional connectivity investigations (Basile et al., 2021; Kumar et al., 2017, 2022). By isolating pulvinar-cortical from broader thalamocortical connectivity, our analysis was able to provide additional insights into the spatial organization of its connectivity with different cortical networks, highlighting the pulvinar's remarkable functional diversity and complexity.”

(4) In addition to DWI/DSI and PET, the study also uses fMRI, which allows for functional interaction in time. It may be worth reflecting in the discussion that the observed gradient organization of the pulvinar could have detailed aspects in the temporal domain, which might not be fully captured in the time-averaged embeddings.

We thank the Reviewer for their insightful observation. The authors recognize that the exploration of brain temporal dynamics is a compelling area of research due to its extensive correlation with multiple hierarchical aspects of brain information processing. Examining the functional organization of the pulvinar complex lies beyond the scope of the present work and will be subject of further investigation. On the other hand, it is possible that certain aspects of the spatial organization of pulvinar connectivity may be influenced by temporal dynamics of cortico-thalamic information processing. Intrinsic timescales have been consistently showed to progressively increase from unimodal to multimodal associative cortical regions. Furthermore, cortico-thalamic connectivity in matrix-rich regions has been correlated with cortical time scales.

To address this point, we have added the following lines to the Discussion section:

“In this context, it could be hypothesized that the observed gradient organization of the pulvinar may also exhibit specific patterns in the temporal domain. Indeed, multiple investigations have linked the temporal dynamics of cortical regions to different aspects of information processing (Rossi-Pool et al., 2021; Soltani et al., 2021). Notably, intrinsic neural timescales of functional activity have been associated with the functional specialization and gradient organization of the cerebral cortex (Golesorkhi et al., 2021), with shorter timescales in unimodal sensory regions and longer ones in transmodal networks (Ito et al., 2020; Murray et al., 2014). Moreover, thalamocortical connectivity has been showed to correlate with these patterns of intrinsic time scale (Müller et al., 2020). In addition, modulatory neurotransmitters such as serotonin and dopamine have been demonstrated to play a significant role in modulating functional cortical dynamics across different timescales (Hansen, Shafiei, et al., 2022; Luppi et al., 2023). Exploring how the spatial organization of the pulvinar relates to temporal dynamics and timescale modulation could provide valuable insights and represents a promising avenue for future investigations.”

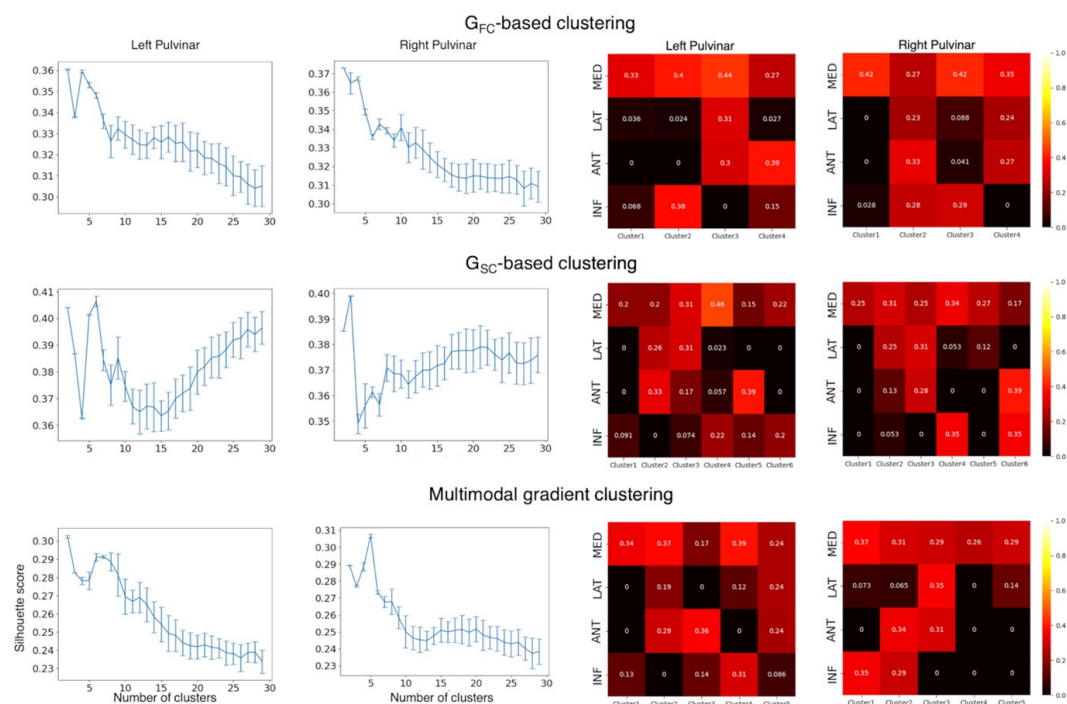
(5) The K-means clustering (Supplementary Figure 1) used has limitations, particularly with respect to the structure of the data. Another aspect is the reproducibility of the model-order selection. Did the reliability and reproducibility assessment produce a similar number of clusters with the LEMON data as with the HCP data?

We acknowledge the limitations of k-means clustering, particularly regarding the stability and reproducibility of the model order. To address the concerns, we iteratively ran the clustering algorithm 50 times on bootstrap resamples to enhance the stability of the silhouette score estimates. In addition, we have now replicated the analysis on the secondary dataset, as suggested by the Reviewer (Author response image 2). The Silhouette plots show similar number of clusters between the two different datasets for functional connectivity gradients, with minor differences observed in the results for structural connectivity gradients and multimodal gradient clustering. Notably, we did not find high a high degree of similarity between the results of gradient clustering and histologically defined nuclei, further underscoring the distinct organizational patterns identified through our analysis.

This reinforces the relevance of using gradient-based approaches to reveal insights into the functional and structural organization of the pulvinar complex that may not align strictly with discrete, histologically defined subdivisions.

Author response image 2.

K-means clustering of pulvinar gradients on the secondary dataset (LEMON) and their correspondence with histological pulvinar nuclei. Panels on the left show the silhouette plots for left and right pulvinar clustering solutions; error bars are standard error calculated across 50 resamples. Panels on the right show matrix plots of Dice similarity coefficients for pulvinar clusters against histological nuclei (AAL3 atlas). INF: inferior; ANT: anterior; LAT: lateral; MED: medial.



(6) The pulvinar correlates of the unimodal-transmodal cortical gradient (Figure 4) show an association with almost the entire brain (Figure 4C, violin plot). It would be interesting to back this association with known anatomical connectivity studies in animals that show connections to these network areas. To my limited knowledge, I am not aware of pulvinar tracer studies showing such extensive connectivity across the entire cortex.

As our structural connectivity estimates are based on tractography, they are subject to the known limitation of potentially overestimating anatomical connectivity. A technical clarification is warranted: since structural connectivity is grouped by networks, it is strongly influenced by connections to specific cortical regions within each network. This explains the uneven and asymmetric distribution of structural gradient-weighted connectivity observed in our results and does not imply widespread connectivity across the entire cortex.

Nonetheless, structural connectivity of the pulvinar to cortical regions in primates encompasses a remarkably broad array of cortical areas, including predominantly occipital (Adams et al., 2000; Benevento, 1976; Casanova et al., 1989), temporal (Berman & Wurtz, 2010; Gattass et al., 2018; Homman-Ludiye et al., 2020) and parietal cortices (Asanuma et al., 1985; Baleydi & Morel, 1992). Additionally, to a more limited extent, connections to the cingulate

gyrus, and portions of the lateral prefrontal cortex have also been documented (Baleydier & Mauguire, 1985; Baleydier & Mauguire, 1987). These connectivity patterns are in line with prior accounts of structural connectivity of the human pulvinar (Arcaro et al., 2015; Basile et al., 2021; Leh et al., 2008; Tamietto et al., 2012), and with the patterns identified in our work (Author response image 1). Such findings provide further validation of the structural connectivity profiles explored in the present study.

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